LM01 Derivative Instrument and Derivative Market Features

Version 1.0

1. Introduction

Cash markets refers to markets in which financial assets like equities, fixed income securities, currencies and commodities are exchanged at current prices. Cash markets are also known as spot markets because the transactions are settled on the spot.

In contrast, derivatives involve the future exchange of cash flows whose value is derived from or based on an underlying value.

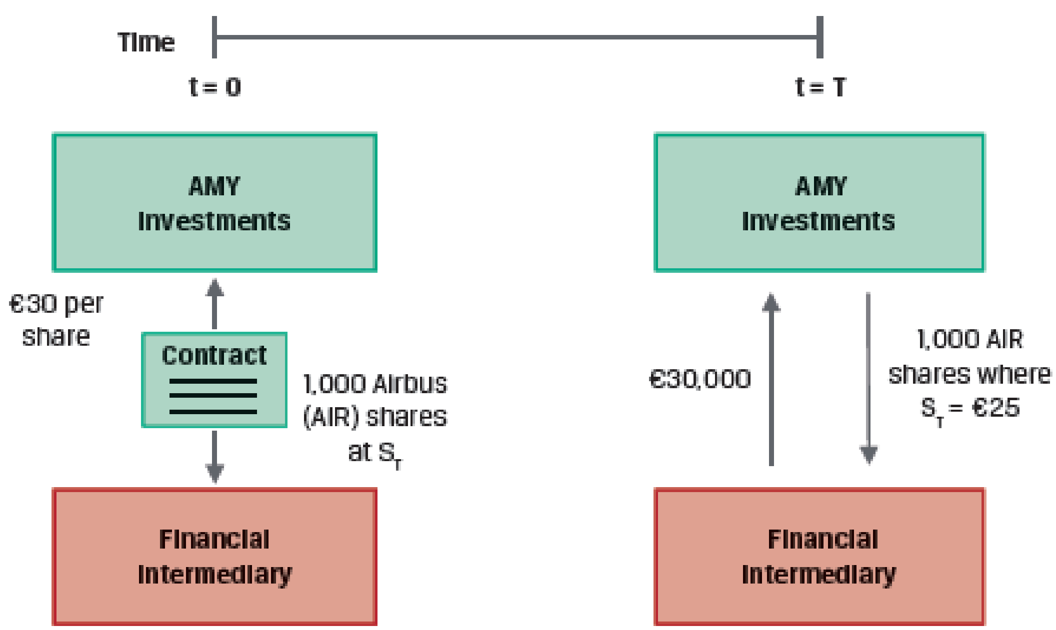
This learning module covers what is a derivative, the different types of derivatives, and the basic features of a derivatives market.

2. Derivative Features

Definition and Features of a Derivative

A derivative is a financial instrument that derives its value from the performance of an underlying asset. In simple terms, a derivative is a legal contract between a buyer and a seller, entered into today, regarding a transaction that will be fulfilled at a specified time in the future. This legal contract is based on an underlying asset.

For example, exhibit 1 from the curriculum shows a 'forward contract' which is a type of derivative.



In this example:

- AMY Investments agrees today (t = 0) to deliver 1,000 shares of Airbus (AIR) at a fixed price of €30 per share on a future date (t = T), which in the example is six months.
- The forward contract allows AMY to transfer the price risk of the underlying to the

counterparty.

- If the spot price of AIR is €25 after six months, AMY will still receive €30,000 from the counterparty in exchange for 1,000 AIR shares (which are now worth just €25,000). Alternatively, AMY could simply settle with the intermediary the €5,000 difference in cash.

A **derivative contract** is a legal agreement between counterparties that defines the rights of each party involved. There are two parties participating in the contract: a buyer and a seller.

- Long: Buyer of the derivative is said to be long on the position. He has the right to buy the underlying according to the conditions mentioned in the contract.
- Short: Seller of the derivative is said to be short. *Remember "s" is for short and seller.*

A **stand-alone derivative** is a distinct derivative contract such as a derivative on a stock or bond. In contrast, an **embedded derivative** is a derivative within an underlying, such as a callable, puttable, or convertible bond.

Exhibit 2 from the curriculum provides a sample forward contract term sheet for the AMY example.

Contract Type: Firm commitment or contingent right to exchange future cash flows	**Forward Transaction Term Sheet**	
Maturity: Final date upon which payment or settlement occurs	**Start Date:**	[Spot start]
	Maturity Date:	[Six months from Start Date]
Counterparties: Legal entities entering the derivative contract	**Forward Purchaser:**	[Financial Intermediary]
	Forward Seller:	AMY Investments
Underlying: Reference asset or variable used as source for contract value **Contract Size:** Amount(s) used for calculation to price and value the derivative	**Forward Delivery:**	1,000 shares of Airbus (AIR) common stock traded on the Frankfurt Stock Exchange
Underlying Price: Pre-agreed price for commitment or contingent claim settlement	**Forward Price:**	€30 per share
Contract Details	**Business Days:**	Frankfurt
	Documentation:	ISDA Agreement and credit terms acceptable to both parties

Derivatives can be divided into two types:

- **Firm commitments**: Both long and short parties have an obligation to complete the transaction. Example: forwards, futures, swaps
- **Contingent claims**: Seller has the obligation. The buyer has a right but no obligation to complete the transaction. Example: Options, credit derivatives.

Derivative markets expand the set of opportunities available to market participants beyond the cash market to create or modify exposure to an underlying. For example,

- Investors can sell short to profit from a decline in the underlying's value.
- Investors can use derivatives to diversify their portfolios.
- Investors can offset financial market exposure associated with a commercial transaction.
- Investors can create large exposures to an underlying with a relatively small cash outlay.
- Derivatives are typically more liquid and have lower transaction costs than the underlying.

Investors can use derivatives for **hedging**. This involves offsetting or neutralizing an existing or anticipated exposure to an underlying.

3. Derivative Underlyings

The most common derivative underlyings include equities, fixed income and interest rates, currencies, commodities, and credit.

Equities

- Equity derivatives can be based on an individual stock, a group of stocks, or a stock index.
- Options on individual stocks are often used by companies as compensation for their employees. Options encourage employees to work to increase the equity value of the company.
- Index swaps, also known as equity swaps, allow investors to pay the return on one stock index in exchange for the return on another index or interest rate.
- Derivatives are also available based upon the realized volatility of equity index prices over a certain period. This allows investors to manage the magnitude of price changes separately from the direction of the change.

Fixed-Income Instruments

- Derivatives based on bonds are widely used. Since government issuers have many similar bond issues outstanding, bond futures typically allow more than one bond issue to be delivered to settle the contract.

- Derivatives based on interest rates are also common. Here the underlying is not an asset.
- Interest rate swaps can be used to convert from a fixed to a floating interest rate exposure over a certain period. A market reference rate (MRR) is the most common interest rate underlying used in interest rate swaps. These rates are based on a daily average of observed market transactions, e.g., the Secured Overnight Financing Rate (SOFR) is an overnight cash borrowing rate collateralized by US Treasuries.

Currencies

- Currency derivatives are often used to hedge the exposure of commercial (e.g. exporters and importers) and financial transactions (e.g. an investor holding foreign securities) that arise due to foreign exchange risk.

Commodities

- Commodities can be classified into soft commodities (agricultural products such as cattle and corn) and hard commodities (natural resources such as crude oil and metals).
- Commodity derivatives are often used to hedge the price risk of commodities. For example, an airline may purchase oil futures to hedge against higher fuel costs in the future.

Credit

- Credit derivatives are based on the default risk of a single issuer or a group of issuers in an index.
- Credit default swaps (CDS) allow an investor to manage the risk of loss from borrower default separately from the bond market.

Other

- Derivatives can also be based on several different types of underlying such as weather, cryptocurrencies, longevity etc.

4. Derivative Markets

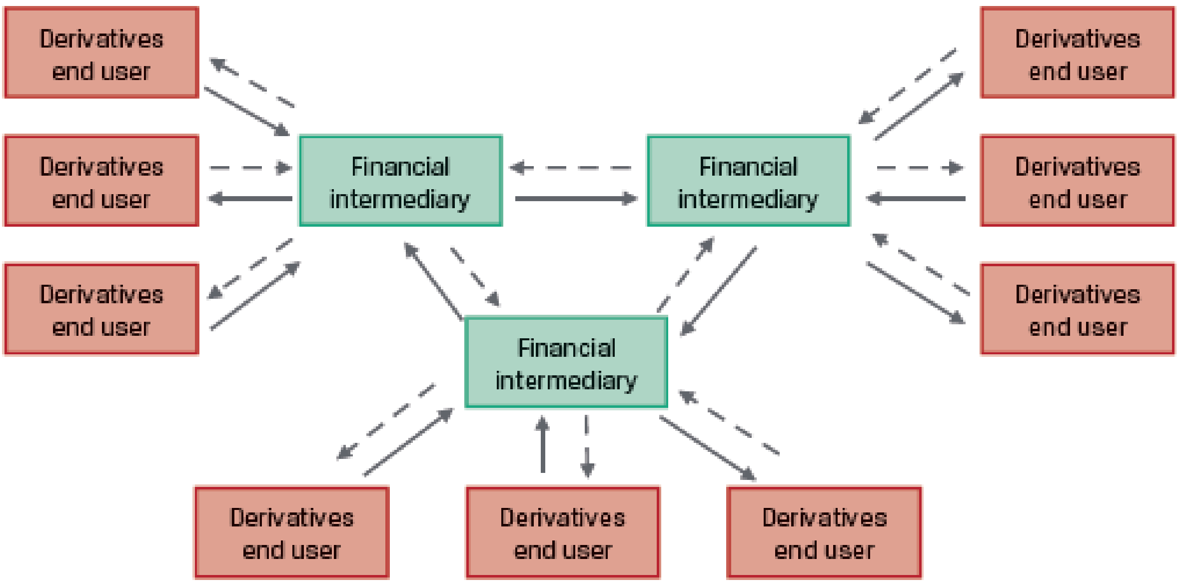
In this section, we look at the characteristics of over-the-counter (OTC) and exchange-traded derivatives (ETD) markets.

Over-the-Counter (OTC) Derivative Markets

- OTC derivatives market involve contracts between derivatives end users and dealers (financial intermediaries such as commercial banks and investment banks).
- The terms of the contracts can be customized as per the end user' needs. This flexibility is important to end users seeking to hedge a specific exposure based on non-standard terms.
- OTC dealers, also called market makers, typically enter into offsetting transactions with

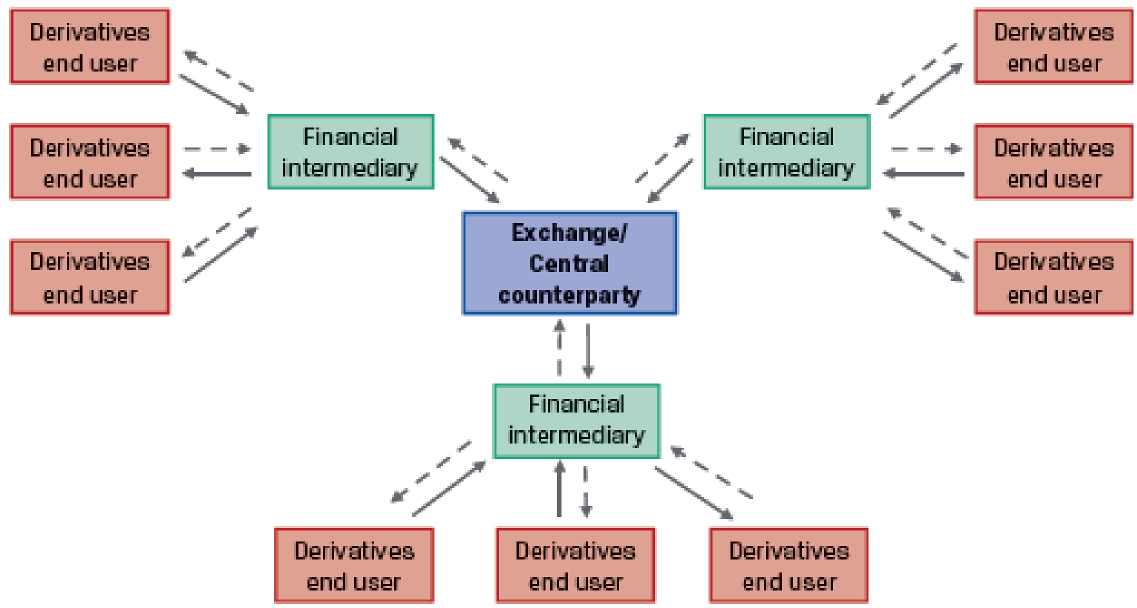
one another to transfer risk to other parties.

- OTC instruments have less transparency and more counterparty risk as compared to ETD.
- Exhibit 4 from the curriculum illustrates the structure of an OTC derivatives market.



Exchange-Traded Derivative (ETD) Markets

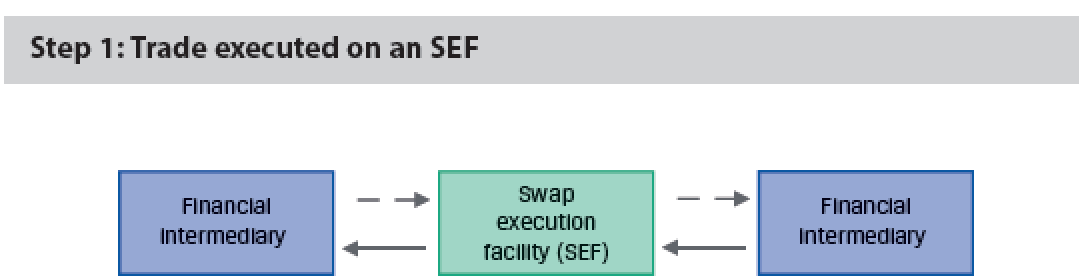
- Exchange-traded derivatives are standardized. The terms and conditions—such as the size of each contract, type, quality, and location of underlying for commodities and maturity date—are set by the exchange. There is no room for customization.
- Standardization leads to a more liquid and transparent market.
- Standardization also facilitates an efficient clearing and settlement process. Clearing is the exchange's process of verifying the execution of a transaction, exchange of payments, and recording the participants. Settlement involves the payment of final amounts and/or delivery of securities or physical commodities between the counterparties.
- To provide a guarantee against counterparty default, derivative exchanges require both the long and short party to post collateral when a trade is initiated. Additional collateral may be required based on the price movements of the underlying during the life of the trade.
- ETD markets are transparent - full information on all transactions is disclosed to the exchanges and national regulators.
- Exhibit 6 from the curriculum illustrates the structure of ETD markets.



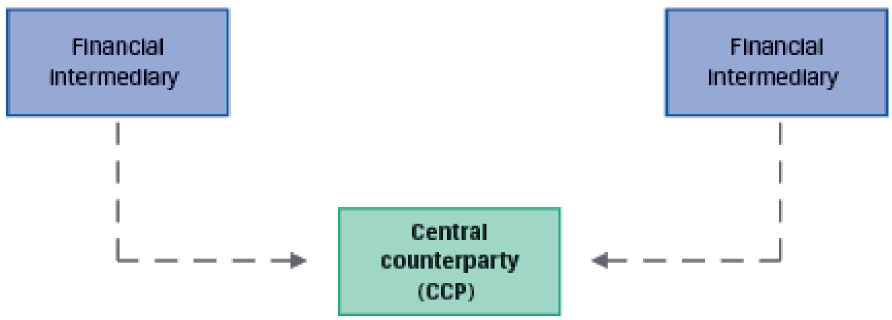
Central Clearing

- After the 2008 global financial crisis, global regulatory authorities instituted a central clearing mandate for most OTC derivatives. This mandate requires that a central counterparty (CCP) assume the credit risk between derivative counterparties. The CCP provides clearing and settlement.
- End users have the flexibility to customize contracts when facing financial intermediaries. While the management of credit risk, clearing, and settlement of transactions between financial intermediaries now occurs in a way similar to ETD markets.
- Although useful, this systemic credit risk transfer from financial intermediaries to CCPs can lead to centralization and concentration of risks. Proper safeguards must be in place to avoid excessive risk being held in CCPs.

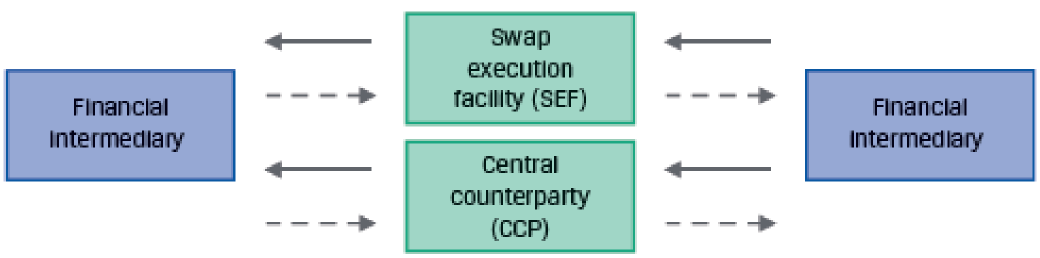
Exhibit 7 shows the 3-step central clearing process for interest rate swaps. The same process also applies to other derivative instruments.



Step 2: SEF trade information submitted to CCP



Step 3: CCP replaces (novates) existing trade, acting as new counterparty to both financial intermediaries



In Step 1, a derivatives trade is executed on a swap execution facility (SEF), which is a swap trading platform that multiple dealers can access.

In Step 2, the original transaction details are shared with the CCP.

In Step 3, the CCP replaces the existing trade. The initial SEF contract is replaced by identical trades facing the CCP in this **novation process**. The CCP acts as a counterparty to both financial intermediaries, eliminating bilateral counterparty credit risk and providing clearing and settlement services.

Summary

LO. Define a derivative and describe basic features of a derivative instrument.

A derivative is a financial instrument that derives its value from the performance of an underlying asset.

Derivatives can be divided into two types:

- Firm commitments: Both parties have an obligation to complete the transaction. Example: Forwards, futures, swaps.
- Contingent claims: Seller has the obligation. The buyer has a right but no obligation to complete the transaction. Example: Options, credit derivatives.

The most common derivative underlyings include equities, fixed income and interest rates, currencies, commodities, and credit.

Derivative markets expand the set of opportunities available to market participants beyond the cash market to create or modify exposure to an underlying.

LO. Describe the basic features of derivative markets, and contrast over-the-counter and exchange-traded derivative markets.

OTC derivatives market involve contracts between derivatives end users and financial intermediaries. The terms of the contracts can be customized as per the end user' needs.

ETDs are standardized contracts that are traded on an exchange. They require collateral on deposit to protect against counterparty default.

For derivatives that are centrally cleared, a central counterparty (CCP) assumes the counterparty credit risk of the financial intermediaries and provides clearing and settlement services.

LM02 Forward Commitment and Contingent Claim Features and Instruments

Version 1.0

1. Introduction

A derivative is a financial instrument that derives its value from the performance of an underlying.

This learning module covers the basic features and valuation of the following derivative instruments:

- Firm commitments: Forwards, futures, and swaps
- Contingent claims: Options and credit derivatives

2. Forwards

A firm commitment is a contract that requires both the long and short parties to engage in a transaction at a later point in time (the expiration) on terms agreed upon today. The parties establish the identity and quantity of the underlying, the manner in which the contract will be executed or settled when it expires, and the fixed price at which the underlying will be exchanged. Both the parties – the buyer and the seller - have an obligation to engage in the transaction at a future date in a firm commitment.

We will look at the three types of firm commitments: forward contracts, futures contracts, and swaps.

Forward Contracts

A forward contract is an over-the-counter derivative contract in which two parties agree to exchange a specific quantity of an underlying asset at a later date at a fixed price they agree on when the contract is signed. It is a customized and private contract between two parties.

Terms of a forward contract

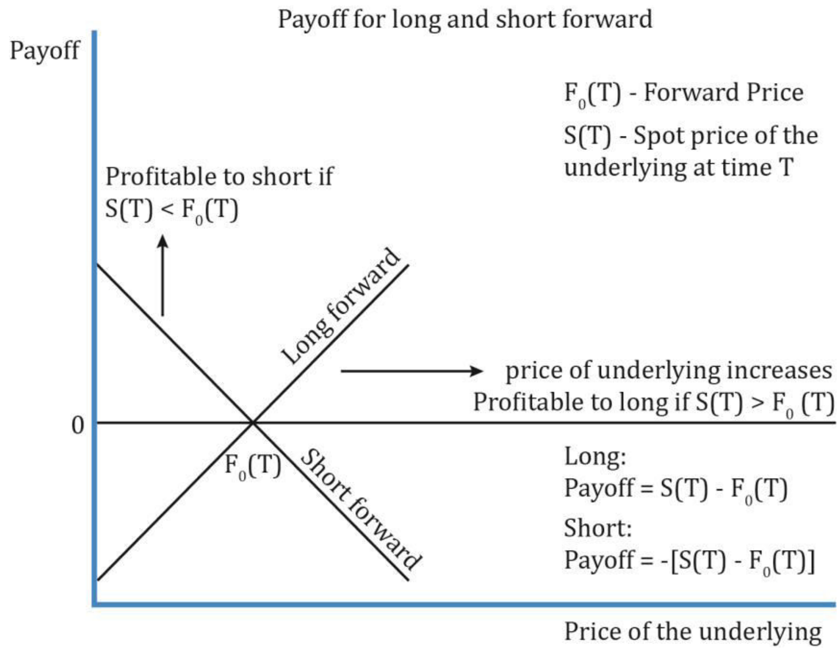
- Price.
- Where the asset will be delivered.
- Identity of the underlying.
- Number of units or quantity of the underlying. For example, if the underlying is gold and the price is fixed per gram, the number of grams to be delivered must be specified.
- It is a customized contract between two parties and not traded over the exchange.

Risk of a forward contract

- The long hopes the underlying will increase in value while the short hopes the asset will decrease in value. One of the two will happen and whoever owes money may default.
- With forward contracts, no money is exchanged at the start of the contract. Further, the value of the contract is zero at initiation.

Payoff for long and short:

The diagram below illustrates the payoff diagram for long and short in a forward contract:



Notation:

- S_0 = price of the underlying at time 0; S_T = price of the underlying at time T
- F_T = forward price fixed at inception at time t = 0
- T = when the forward contract expires

The long party benefits if the spot price increases and goes above the forward price. The short party benefits if the spot price decreases and goes below the forward price.

Example

Whizz wants to sell 500 shares of beverage maker FTC to Fizz at $50 per share after 180 days. What happens if the market price of FTC at expiry is $50, $60, $70 or $40?

Solution:

Fizz is the long party and Whizz is the short party.

At expiry from Fizz's perspective (long) when market price is:

$50: payoff is 0 and no one gains or loses

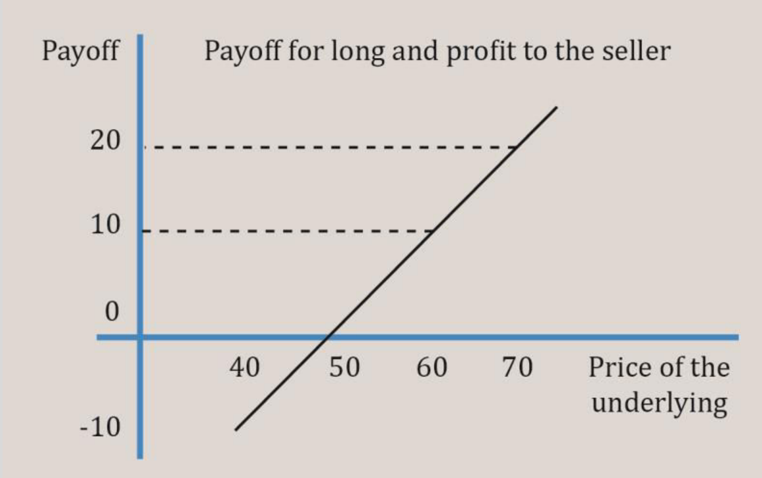
$60: long gains by $10

$70: long gains by $20

$40: long loses by $10 as he can buy the asset cheaper by $10 directly from the market instead of paying $50 to Whizz

From Whizz's perspective, it is exactly the opposite to that of Fizz. When the share price increases to $70, he loses $20 as he can sell it in market for $70 but is selling for only $50 to Fizz.

The payoff for the long is depicted diagrammatically below:



3. Futures

A futures contract is a standardized derivative contract created and traded on a futures exchange such as the Chicago Mercantile Exchange (CME). In a futures contract, two parties agree to exchange a specific quantity of the underlying asset at an agreed-upon price at a later date. The buyer agrees to purchase the underlying asset from the other party, the seller. The agreed-upon price is called the **futures price.**

There are some similarities with a forward contract: two parties agreeing on a contract, an underlying asset, a fixed price called the futures price, a future expiry date, etc. But the following characteristics differentiate futures from a forward:

- The contracts are standardized.
- They are traded on a futures exchange.
- The fixed price is called futures price and is denoted by *f.* (Forward prices are usually represented by *F.*)
- The biggest difference is that gains or losses are settled on a daily basis by the exchange through its clearinghouse. This process is called **mark to market(MTM).**
- Settlement price is the average of final futures trades and is determined by the clearinghouse.
- The futures price converges to a spot price at expiration.
- At expiry: the short delivers the asset and the long pays the spot price.

Let us take an example. Assume Ann enters into a contract to buy 100 grams of gold at $55 per gram after 90 days. The futures price is $55. At the end of day 1, the futures price is $58. There is a gain of $3. So, $300 ($3 per gram x 100 grams) is credited to Ann's account. This is called marking to market. The account maintained by Ann is called the margin account.

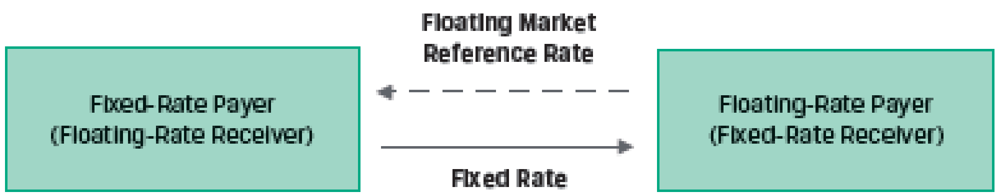
Initial margin is the amount that must be deposited in a futures account before a trade is made. Associated with the initial margin is another figure called the **maintenance margin** which is the amount of money each participant must maintain in their accounts after the trade is initiated. If the margin balance falls below the maintenance margin additional funds must be deposited to bring the balance up to the initial margin amount.

To limit the potential losses from counterparty default, some futures contracts may limit daily price changes. These **price limits** set a band relative to the previous day's settlement price within which all trades must take place. If price reaches the band limit, trading stops until two parties agree on a trade at a price within the prescribed range. In some cases, exchanges may use a **circuit breaker –** which pause intraday trading for a brief period, if a price limit is reached.

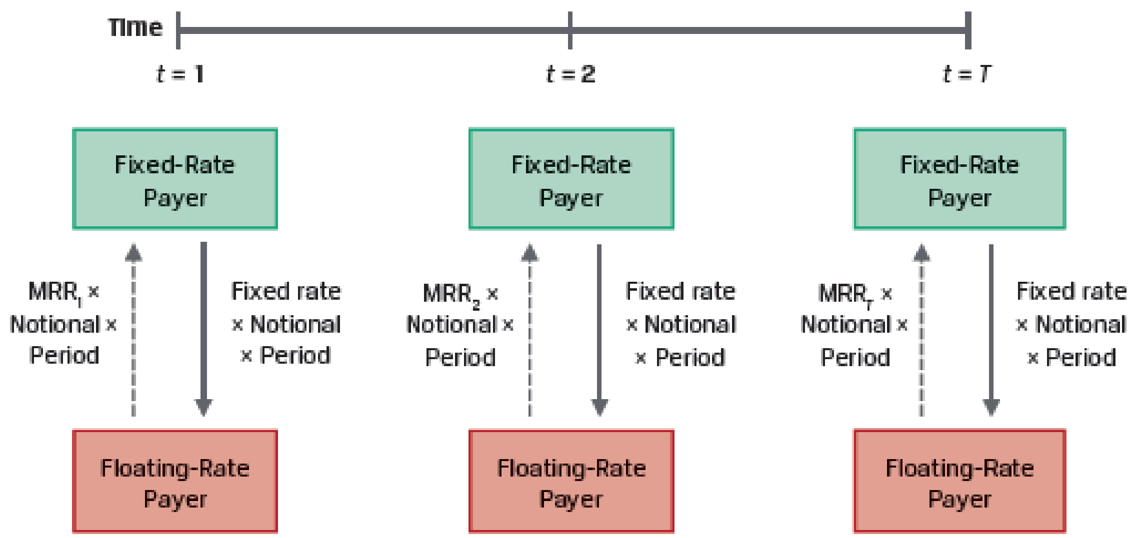
4. Swaps

A swap is a firm commitment between two counterparties to exchange a series of cash flows in the future. Typically, one cash flow stream is variable based on a market reference rate that resets each period; and the other cash flow stream is fixed or may vary based on a different underlying asset or rate.

Interest rate swaps are the most common type of swaps. Here, a fixed exchange rate is exchanged for a floating interest rate. The counterparty paying variable cash flows is called the **floating rate payer**; and the counterparty paying fixed cash flows is called the **fixed rate payer**. This is illustrated in Exhibit 3 of the curriculum.



The market reference rate (MRR) paid by the floating-rate payer resets every period, while the fixed rate (also called the **swap rate**) remains constant, as shown in Exhibit 4.

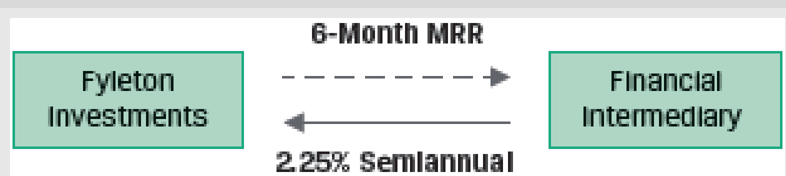


The payments are based on an agreed upon notional principal. The notional principal is usually not exchanged, but only used to calculate the payments. The payments are typically netted. A fixed rate payer receives a net payment if the MRR exceeds the fixed rate for a given period. Swaps allow investment managers to change portfolio duration without trading bonds.

Example: Fyleton Investments Swap

(*This is Example 4 from the curriculum.*)

Fyleton Investments has entered a five-year, receive-fixed GBP200 million interest rate swap with a financial intermediary to increase the duration of its fixed-income portfolio. Under terms of the swap, Fyleton has agreed to receive a semiannual GBP fixed rate of 2.25% and pay six-month MRR.



Calculate the first swap cash flow exchange if six-month MRR is set at 1.95%.

- The financial intermediary owes Fyleton a fixed cash flow payment of GBP2,250,000 (= GBP200 million × 0.0225/2).
- Fyleton owes the financial intermediary a floating cash flow payment of GBP1,950,000 (= GBP 200 million × 0.0195/2).
- The fixed and floating payments are netted against one another, and the net result is that the financial intermediary pays Fyleton GBP300,000 (= GBP2,250,000 – GBP1,950,000).

5. Options

We now move to the other major category of derivative instruments called contingent claims. The holder of a contingent claim has the right, but not the obligation to make a final payment contingent on the performance of the underlying. Options are the most common type of contingent claims.

Options

An option is a derivative contract in which one party, the buyer, pays a sum of money to the other party, the seller or writer, and receives the right to either buy or sell an underlying asset at a fixed price either on a specific expiration date or at any time prior to the expiration date. Options trade on exchanges, or they can be customized in the OTC market.

The buyer/holder of an option is said to be long.

The seller/writer of an option is said to be short.

There are two types of options based on when they can be exercised:

- European option: This type of option can be exercised only on the expiration date.
- American option: This type of option can be exercised on or any time before the option's expiration date.

There are two types of options based on the purpose it serves:

- Call option: Gives the buyer the right to buy the underlying asset at a given price on a specified expiration date. The seller of the option has an obligation to sell the underlying asset.
- Put option: Gives the buyer the right to sell the underlying asset at a given price on a specified expiration date. The seller of the option has an obligation to buy the underlying asset.

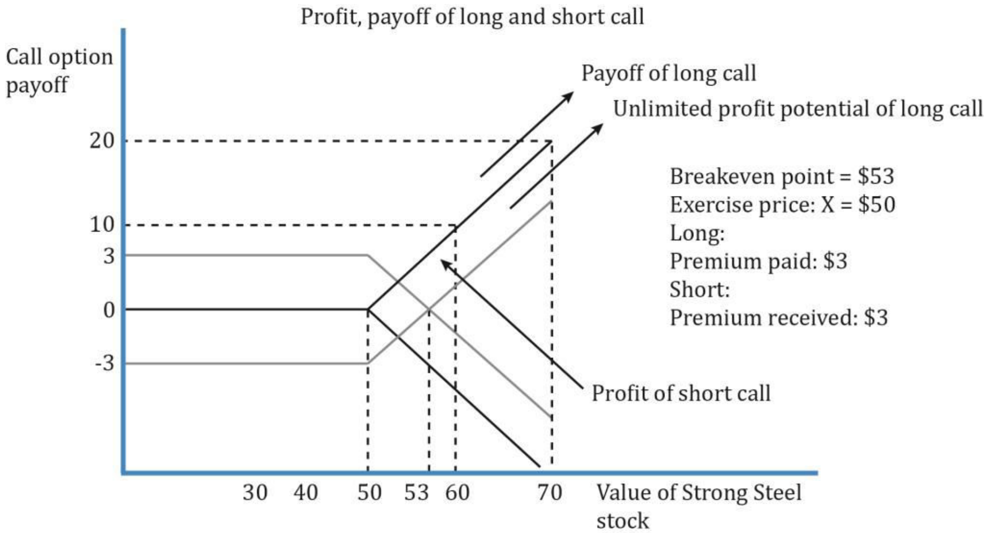
Assume there are two parties: A and B. A is the seller, writer, or the short party. B is the buyer or the long party. A and B sign a contract, according to which B has the right to buy one share of Strong Steel Inc. for \$50 after six months.

In our example, B has bought the right to buy, which is called a call option. The right to sell is called a put option. If B has the right to buy a share (exercise the option) of Strong Steel Inc. anytime between now and six months, then it is an American-style option. But if he can exercise the right only at expiration, then it is a European-style option. \$50, the price fixed at which the underlying share can be purchased, was fixed at inception and is called the strike price or exercise price.

B bought the right to buy the share at expiration from A. So, B has to pay A, a sum of money called the option premium for holding this right without an obligation to purchase the share.

The call premium B paid is \$3. An investor would buy a call option if he believes the value of the underlying would increase.

The diagram below shows the call option payoff and profit for both a buyers and sellers perceptive.



<u>Call option buyer and seller payoff at expiration:</u>

$c_T = Max\ (0, S_T - X)$

where:
S_T = *stock's price*
X = *exercise price*

For example, suppose you buy a call option with an exercise price of 30 and an expiration of three months for a premium of 1.00 when the stock is trading at 25. At expiration, consider the outcomes when the stock's price is 25, 30, or 35. The buyer's payoffs would be:

For S_T = 25, payoff = $c_T = Max(0, S_T - X) = Max(0, 25 - 30)$ = Max(0, −5) = 0.
For S_T = 30, payoff = $c_T = Max(0, S_T - X) = Max(0, 30 - 30)$ = Max(0, 0) = 0.
For S_T = 35, payoff = $c_T = Max(0, S_T - X) = Max(0, 35 - 30)$ = Max(0, 5) = 5.

To the seller, who received the premium at the start, the payoff is:
$-c_T = -Max(0, S_T - X)$

At expiration, the call seller's payoffs are:

For S_T = 25, payoff = $-c_T = -Max(0, S_T - X) = -Max(0, 25 - 30) = 0$.
For S_T = 30, payoff = $-c_T = -Max(0, S_T - X) = -Max(0, 30 - 30) = 0$.
For S_T = 35, payoff = $-c_T = -Max(0, S_T - X) = -Max(0, 35 - 30) = -5$.

The call buyer's profit would be:

Profit = $Max(0, S_T - X) - c_0$

where:

c_0 = option premium

For S_T = 25, profit = $Max(0,S_T - X) - c_0 = Max(0,25 - 30) - 1.00 = - 1$.
For S_T = 30, profit = $Max(0,S_T - X) - c_0 = Max(0,30 - 30) - 1.00 = - 1.00$.
For S_T = 35, profit = $Max(0,S_T - X) - c_0 = Max(0,35 - 30) - 1.00 = 4.00$.

The call seller's profit is:

$\Pi = -Max(0,S_T - X) + c_0$

At expiration, the call seller's profit for each underlying price at expiration are:

For S_T = 25, profit = $-Max(0,S_T - X) + c_0 = -Max(0,25 - 30) + 1.00 = 1.00$.
For S_T = 30, profit = $-Max(0,S_T - X) + c_0 = -Max(0,30 - 30) + 1.00 = 1.00$.
For S_T = 35, profit = $-Max(0,S_T - X) + c_0 = -Max(0,35 - 30) + 1.00 = -4.00$.

Put option buyer and seller payoff at expiration:

An investor would buy a put option if he believes the value of the underlying would decrease. If it decreases by expiration date, the investor has a right to exercise the option to sell the underlying at the exercise price, which will be greater than the then market price.

The payoff to the put holder is:

$p_T = Max(0,X - S_T)$

The put buyer's profit would be:

$\Pi = Max(0,X - S_T) - p_0$

where: p_0 is option premium.

The payoff for the seller is:

$-p_T = -Max(0,X - S_T)$

The put seller's profit would be:

$\Pi = -Max(0,X - S_T) + p_0$

Intrinsic value and time value of options

A call option is **in-the-money** when the price of the underlying is higher than the exercise price (i.e. $S_t > X$). It is **out-of-the-money** if ($S_t < X$) and **at-the-money** if ($S_t = X$).

Example: On 15 November, a 90-day European-style call option with an exercise price of \$50.00 is priced at \$4.60. The underlying is trading at \$53.00. Hence, T = 90/365, S_0 = 53.00, X = 50.00, and c_0 = 4.60.

The option price has two components: 1) an exercise value component, also called intrinsic value, and 2) a time value component. Hence, we can say: c = exercise value + time value.

- For a call option, the exercise value component is S – X if S > X. In our example, the

exercise value is 53.00 – 50.00 = 3.00.
- The time value component is equal to the option price minus that exercise value: 4.60 – 3.00 = 1.60.

The longer the time to expiration the higher the time value. The time value declines as time passes and becomes equal to zero at option expiration. At expiration, call and put values are equal to their intrinsic value or exercise value.

Counterparty credit risk

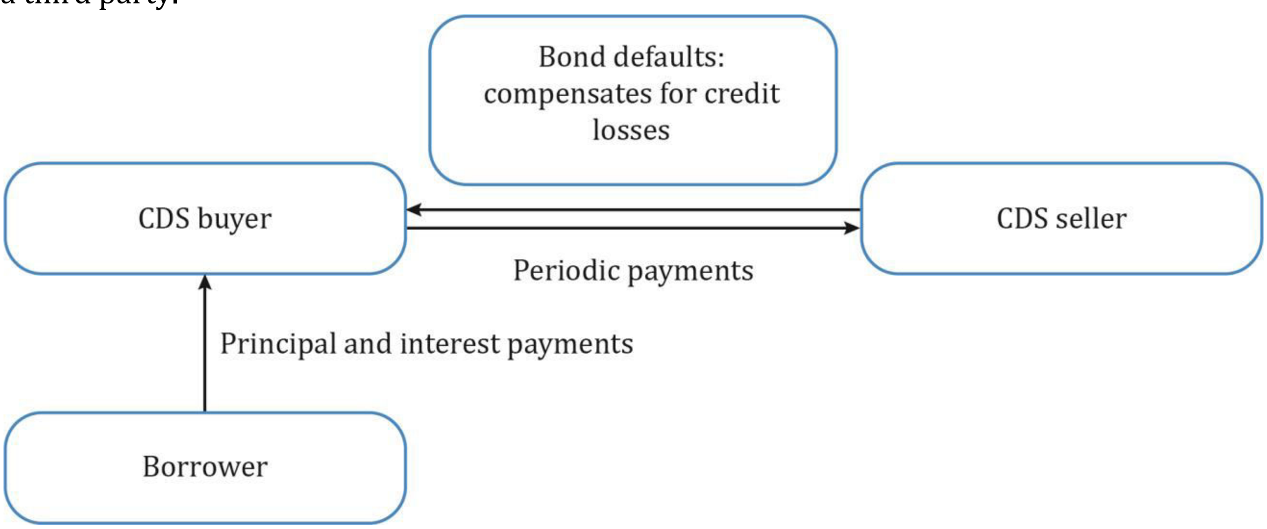
Options have one-sided counterparty credit risk. The option seller has no credit exposure to the option buyer after the premium is paid. However, the option buyer faces the counterparty credit risk of the option seller equal to the option payoff at maturity.

6. Credit Derivatives

A credit derivative is a class of derivative contracts between two parties, a credit protection buyer and a credit protection seller, in which the latter provides protection to the former against a specific credit loss. The main type of credit derivative is a credit default swap.

Credit Default Swap

A credit default swap is a derivative contract between two parties, a credit protection buyer and a credit protection seller, in which the buyer makes a series of cash payments to the seller and receives a promise of compensation for credit losses resulting from the default of a third party.



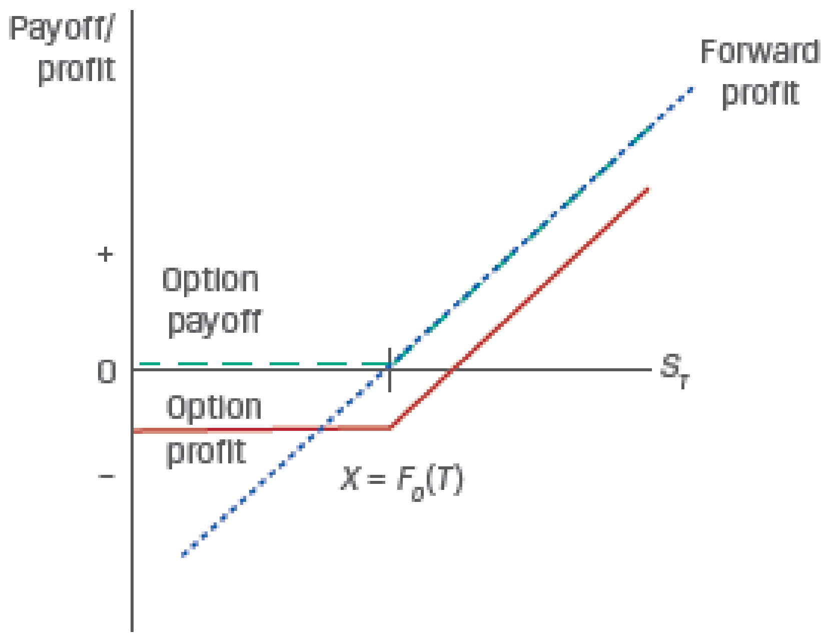
- A CDS is a form of insurance.
- The objective of a CDS is to buy protection to cover the loss of par value of the bond, if a credit event occurs. A credit event is like bankruptcy, restructuring, or failure to pay that impairs the borrower's ability to make timely payments.
- The CDS transfers the credit risk of the borrower from one party (protection buyer) to another (protection seller).

- **Credit migration**: If the borrower's credit spread widens (i.e. credit quality worsens) the CDS buyer has a MTM gain and the CDS seller has a MTM loss. The CDS MTM change is calculated as: Δ CDS spread x Notional x $EffDur_{CDS}$
- **Credit event**: If the borrower defaults, the CDS seller pays the CDS buyer and the CDS contract terminates. The contingent payment is equal to the issuer Loss given default (LGD)% x the notional amount.

7. Forward commitments vs contingent claims

Market participants frequently create similar exposures to an underlying using different derivative instrument types. For example, both a long forward position and a call option position will benefit from an increase in the price of the underlying.

Exhibit 11 from the curriculum contrasts the payoff profiles of these two positions.



The profit for the forward contract is simply: $S_T - F_0(T)$

If the exercise price of the call option is set to $F_0(T)$, the profit for the call option is:

$\Pi = \max[0, S_T - F_0(T)] - c_0$.

Based on this, we can draw the following conclusions:

- $S_T - F_0(T) > -c_0$ Forward profit exceeds option profit
- $S_T - F_0(T) = -c_0$ Forward profit equals call option profit
- $S_T - F_0(T) < -c_0$ Option profit exceeds forward profit

When the stock price moves up the profits are similar. But when the stock price moves down, the call option provides downside protection in exchange for paying a premium.

Summary

LO: Define forward contracts, futures contracts, swaps, options (calls and puts), and credit derivatives and compare their basic characteristics.

Forward contract is an over-the-counter derivative contract in which two parties agree to exchange a specific quantity of an underlying asset on a later date, at a fixed price they agree on when the contract is signed. It is a customized and private contract between two parties.

A futures contract is similar to a forward contract in that two parties agree to exchange a specific quantity of the underlying asset at an agreed-upon price at a later date. However, a futures contract is a standardized derivative contract, created and traded on a futures exchange, with daily settlement of gains and losses.

A swap is a firm is a firm commitment between two counterparties to exchange a series of cash flows in the future. Typically, one cash flow stream is variable based on a market reference rate that resets each period; and the other cash flow stream is fixed or may vary based on a different underlying asset or rate.

A call option gives the buyer the right to buy the underlying asset at a given price on a specified expiration date. The seller of the option has an obligation to sell the underlying asset.

A put option gives the buyer the right to sell the underlying asset at a given price on a specified expiration date. The seller of the option has an obligation to buy the underlying asset.

Credit derivative is a class of derivative contracts between two parties, a credit protection buyer and a credit protection seller, in which the latter provides protection to the former against a specific credit loss.

LO: Determine the value at expiration and profit from a long or a short position in a call or put option.

The call option buyer's payoffs at expiration is:

$c_T = Max(0, S_T - X)$

To the seller, who received the premium at the start, the payoff is:

$-c_T = -Max(0, S_T - X)$

The call buyer's profit is:

$Max(0, S_T - X) - c_0$

The call seller's profit is:

$\Pi = -Max(0, S_T - X) + c_0$

The payoff to the put holder is:

$p_T = Max(0, X - S_T)$

The put buyer's profit is:
$\Pi = Max(0, X - S_T) - p_0$

The payoff for the seller is:
$-p_T = -Max(0, X - S_T)$

The put seller's profit is:
$\Pi = -Max(0, X - S_T) + p_0$

LO: Contrast firm commitments with contingent claims.

Firm commitments are contracts entered into at one point in time that require both parties to engage in a transaction at a later point in time (the expiration), on terms agreed upon at the start. Examples: forward contracts, futures contracts, and swaps.

The holder of a contingent claim has the right, but not the obligation, to make a final payment, contingent on the performance of the underlying. Example: options and credit derivatives.

LM03 Derivative Benefits, Risks, and Issuer and Investor Uses

Version 1.0

1. Introduction

This learning module covers:

- The benefits and risks of using derivatives
- A comparison of how issuers use derivatives versus how investors use derivatives

2. Derivative Benefits

Some of the benefits of derivatives are listed below:

Risk Allocation, Transfer, and Management

Derivative instruments enable users to allocate, transfer, and/or manage risk without trading an underlying.

In many scenarios, issuers and investors face a timing difference between making an economic decision and being able to transact in a cash market. The curriculum provides the following examples:

Issuers:

- A manufacturer may need to order commodity inputs for its production process in advance of receiving finished-goods orders.
- A retailer may await a shipment of goods priced in a foreign currency before selling domestic currency to make payment.
- An issuer may wish to lock in its future debt costs in advance of the maturity of an outstanding debt issuance.

Investors:

- An investor may seek to capitalize on a market view but lack the necessary cash on hand to transact in the spot market.
- In anticipation of a future stock dividend, debt coupon, or principal repayment, an investor may decide today how it will reinvest the proceeds in the future.

In such scenarios, derivatives can be used to lock in a pre-agreed price for a future date and bridge the timing gap. For example, if an investor has a bullish view on the market but lacks the cash to transact in the spot market, he can purchase a forward contract or a call option on an index to benefit from his view.

Derivatives allow users to create exposure profiles which are unavailable in cash markets. For example, a short call position can be added to an existing long cash position to create a 'covered call' position. A covered call generates a higher return as compared to the original long cash position if the underlying price is stable or slightly higher.

Instructor's Note: This concept is covered in a later reading.

Information Discovery

There are two primary advantages of futures markets:

- Price discovery: Market participants often use futures prices to forecast the direction of cash markets. For example, analysts frequently use interest rate futures markets to gauge investor expectations of a central bank's benchmark interest rate increase or decrease at a future meeting.
- Implied volatility: Implied volatility measures the expected risk of the underlying. With the models such as BSM to price an option, it is possible to determine the implied volatility, and hence the risk.

  Instructor's Note: This concept is covered in a later reading.

Operational Advantages

Some of the major operational advantages associated with derivatives are given below:

- Lower transaction costs than the underlying.
- Greater liquidity than the underlying spot markets.
- Margin requirements and option premiums are low relative to the cost of the underlying
- Easy to take a short position.

Market Efficiency

Derivatives markets offer a less costly way to exploit mispricings in the market due to operational advantages: low transaction costs, easier to take a short position, etc. Therefore, any mispricing is corrected more quickly in the derivatives market than the spot market.

Thus, the existence of derivative markets causes financial markets in general to function more efficiently.

Exhibit 1 from the curriculum summarizes the benefits of derivative instruments.

Purpose	Description
Risk Allocation, Transfer, and Management	Allocate, trade, and/or manage underlying exposure without trading the underlying Create exposures unavailable in cash markets
Information Discovery	Deliver expected price in the future as well as expected risk of underlying
Operational Advantages	Reduced cash outlay, lower transaction costs versus the underlying, increased liquidity and ability to "short"
Market Efficiency	Less costly to exploit arbitrage opportunities or mispricing

3. Derivative Risks

The main risks associated with derivatives are listed below:

Greater Potential for Speculative Use

Derivatives have a high degree of implicit leverage as compared to a similar cash position. This magnifies profits/losses and increases the likelihood of financial distress.

Lack of Transparency

Derivatives increase a portfolio's complexity. They may create an exposure profile that is not well understood by stakeholders.

This risk increases when a combination of derivatives and/or embedded derivatives is used. For example, structured notes are a broad category of securities that combine features of debt instruments with embedded derivatives designed to achieve a particular issuer or investor objective. They often involve greater cost, lower liquidity, and less transparency as compared to an equivalent stand-alone instrument.

Basis Risk

Derivatives are often used to hedge the risk of a commercial or financial exposure. The assumption here is that the derivative instrument will be highly effective in offsetting the price risk of the underlying. However, in some cases, the expected value of a derivative differs unexpectedly from the underlying. This is referred to as 'basis risk'.

Basis risk can occur when a derivative instrument refers to a price or index that is similar to, but does not exactly match, an underlying exposure.

Liquidity Risk

Liquidity risk refers to the divergence in cash flow timing of a derivative versus that of an underlying transaction. For example, future contracts require daily settlement of gains and losses and can give rise to liquidity risk. An investor using futures to hedge an underlying transaction may be unable to meet a margin call due to lack of funds, and may be forced to close out his position.

Counterparty Credit Risk

Derivatives can result in significant counterparty credit exposure. Unlike bond markets where credit exposures are easy to predict based on the notional principal and accrued interest; predicting the credit exposures of derivatives is more challenging. Credit exposures vary based on the type of derivatives. For example, a call option seller has no counterparty credit risk exposure after receiving the premium, but a call option buyer is exposed to counterparty credit risk, which varies depending on the price of the underlying.

Destabilization and Systemic Risk

Derivatives are often blamed to have destabilizing consequences on the financial markets. This is primarily due to the high amount of leverage taken by speculators. If the position turns against them, then they default. This triggers a ripple effect causing their creditors to default, their creditors' creditors to default, and so on. A default by speculators impacts the whole system. For example, the credit crisis of 2008.

Many market reforms such as the central clearing mandate for swaps between financial intermediaries and a central counterparty (CCP), have been put in place to reduce systemic risk.

Exhibit 3 from the curriculum summarizes the key risks associated with derivative instruments and markets.

Risk	**Description**
Greater Potential for Speculative Use	High degree of implicit leverage for some derivative strategies may increase the likelihood of financial distress.
Lack of Transparency	Derivatives add portfolio complexity and may create an exposure profile that is not well understood by stakeholders.
Basis Risk	Potential divergence between the expected value of a derivative instrument versus an underlying or hedged transaction
Liquidity Risk	Potential divergence between the cash flow timing of a derivative instrument versus an underlying or hedged transaction
Counterparty Credit Risk	Derivative instruments often give rise to counterparty credit exposure, resulting from differences in the current price versus the expected future settlement price.
Destabilization and Systemic Risk	Excessive risk taking and use of leverage in derivative markets may contribute to market stress, as in the 2008 financial crisis.

4. Issuer Use of Derivatives

Issuers typically use derivative instruments to offset or hedge market-based underlying exposures that impact their assets, liabilities, and earnings. For example, companies often use FX forward contracts to hedge FX exposures related to import/export transactions.

Issuers also seek hedge accounting, which allows an issuer to offset a hedging instrument (usually a derivative) against a hedged transaction or balance sheet item. This reduces income statement and cash flow volatility.

Hedge types include:

- **Cash flow hedge**: A derivative designated as absorbing the variable cash flow of a floating-rate asset or liability. For example, FX forwards are a cash flow hedge that offset the variability of exchange rates.
- **Fair value hedge**: A derivative used to offset the fluctuation in fair value of an asset or liability. For example, a commodities producer anticipating lower future prices may use forward contracts to sell its inventory and lock-in a price.
- **Net investment hedge**: A net investment hedge occurs when either a foreign currency bond or a derivative such as an FX swap or forward is used to offset the exchange rate risk of the equity of a foreign operation.

To qualify for hedge accounting, the dates, notional amounts, and other contract features of a derivative must closely match those of the underlying transaction. Therefore, issuers are more likely to use OTC derivatives customized to meet their specific needs rather than exchange traded derivatives.

5. Investor Use of Derivatives

Investors use derivatives to:

- replicate a cash market strategy (e.g. purchasing a futures contract on a stock to replicate its cash market performance)
- hedge a fund's value against adverse movements in underlyings (e.g. purchasing put options to protect an equity portfolio from significant losses during an uncertain event)
- modify or add exposures using derivatives which may be unavailable in cash markets (e.g. covered call)

The prospectus of an investment fund usually specifies which derivative instruments may be used within the fund and for what purpose.

Investors are typically less concerned with hedge accounting than issuers. As a result, they tend to trade in standardized and highly liquid exchange-traded derivatives more frequently than issuers.

Summary

LO. Describe benefits and risks of derivative instruments.

Benefits

Purpose	Description
Risk Allocation, Transfer, and Management	Allocate, trade, and/or manage underlying exposure without trading the underlying Create exposures unavailable in cash markets
Information Discovery	Deliver expected price in the future as well as expected risk of underlying
Operational Advantages	Reduced cash outlay, lower transaction costs versus the underlying, increased liquidity and ability to "short"
Market Efficiency	Less costly to exploit arbitrage opportunities or mispricing

Risks

Risk	Description
Greater Potential for Speculative Use	High degree of implicit leverage for some derivative strategies may increase the likelihood of financial distress.
Lack of Transparency	Derivatives add portfolio complexity and may create an exposure profile that is not well understood by stakeholders.
Basis Risk	Potential divergence between the expected value of a derivative instrument versus an underlying or hedged transaction
Liquidity Risk	Potential divergence between the cash flow timing of a derivative instrument versus an underlying or hedged transaction
Counterparty Credit Risk	Derivative instruments often give rise to counterparty credit exposure, resulting from differences in the current price versus the expected future settlement price.
Destabilization and Systemic Risk	Excessive risk taking and use of leverage in derivative markets may contribute to market stress, as in the 2008 financial crisis.

LO. Compare the use of derivatives among issuers and investors.

Issuers typically use derivative instruments to offset or hedge market-based underlying exposures that impact their assets, liabilities, and earnings. Issuers also seek hedge accounting, which allows an issuer to offset a hedging instrument (usually a derivative) against a hedged transaction or balance sheet item. This reduces income statement and cash flow volatility.

Investors use derivatives to replicate a cash market strategy, hedge a fund’s value against adverse movements in the underlying, modify or add exposures using derivatives which may be unavailable in cash markets.

LM04 Arbitrage, Replication, and the Cost of Carry in Pricing Derivatives

Version 1.0

1. Introduction

This learning module covers:

- How the concepts of arbitrage and replication are used in pricing derivatives.
- How the costs or benefits associated with owning an underlying asset affect the forward commitment price.

2. Arbitrage

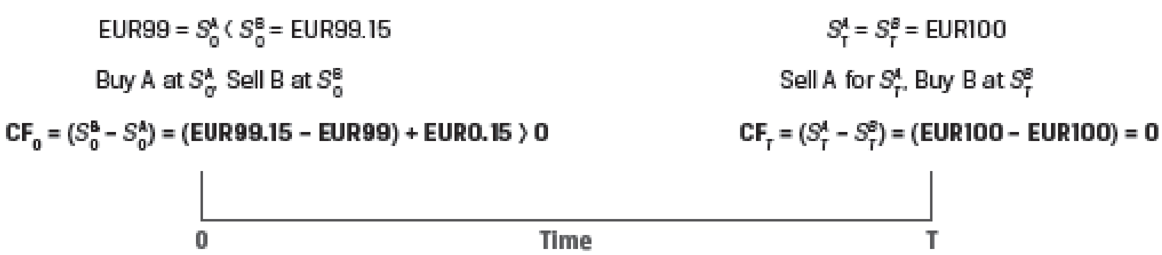
According to the 'law of one price' two identical assets should trade at the same price. An arbitrage opportunity exists if the 'law of one price' does not hold.

In case of derivatives arbitrage opportunities exist when:

1. Two assets with identical future cash flows trade at different prices.
2. An asset with a known future price does not trade at the PV of its future price.

Scenario 1

Exhibit 1 from the curriculum illustrates the first scenario.



Consider two zero-coupon bonds A and B. Both bonds are from the same issuer, have identical features, and will mature on the same future date.

1. Bond A is trading at a price of EUR99 at time t=0.
2. Bond B is trading at a price of EUR99.15 at time t=0.
3. Both bonds have an expected future price of EUR100.

Since two assets with identical future cash flows trade at different prices, the 'law of one price' is violated and an arbitrage opportunity exists.

At time t = 0, an investor can:

- Sell Bond B short to receive EUR99.15 and purchase Bond A for EUR99
- This will give him a net cash inflow at t=0 of EUR0.15

At time t = T, when both bonds mature, the investor:

- Receives EUR100 for Bond A and uses this to buy Bond B for EUR100 to cover the short position.

- This leaves him with a riskless profit of EUR0.15 at time 0.

Other market participants will also exploit this opportunity, selling Bond B, driving its price down, and buying Bond A, driving its price up, until the prices converge.

Scenario 2

The second-type of derivative-related arbitrage opportunity arises when an asset with a known future price does not trade at the PV of its future price.

In an earlier reading on time-value-of-money, we learnt that the future value of a single cash flow can be calculated based on discrete compounding or continuous compounding.

The FV based on discrete compounding is:

$$FV_N = PV(1 + r)^N$$

The FV based on continuous compounding is:

$$FV_T = PVe^{rT}$$

Both approaches can be used for pricing derivatives. The curriculum uses the discrete compounding method for individual underlying assets (e.g. a stock); and the continuous compounding method for underlying assets that represent a portfolio (e.g. an equity index) or where the underlying involves foreign exchange.

The following example from the curriculum demonstrates how an arbitrage profit can be earned in scenario 2.

Example: Spot vs. Discounted Known Future Price of Gold
(*This is based on Example 1 from the curriculum.*)

A company entered into a contract to buy 100 ounces of gold at an agreed-upon price of USD1,792.13 per ounce in three months.

Assume that today's spot gold price (S_0) is USD1,770 per ounce, and the annualized risk-free interest rate (r) is 2%. Also assume that the company can borrow at the risk-free rate and there are no additional costs or benefits associated with gold ownership. Under these conditions demonstrate how the company can generate a riskless profit.

Solution:

At time t = 0,

- The company borrows USD177,000 at 2.0% interest for three months and purchases 100 ounces of gold at today's spot price.
- The company enters into a forward contract today to sell 100 ounces of gold at a price of USD1,792.13 per ounce in three months.

At time t = T (in three months),

- The company delivers 100 ounces of gold under the forward contract and receives USD179,213 (= 100 × USD1,792.13).
- The company repays the loan principal with interest: USD177,878.44 = USD177,000$(1.02)^{0.25}$.
- The company's riskless profit at time T is equal to the difference between the forward sale proceeds and the loan principal and interest: USD1,334.56 = USD179,213 – USD177,878.44.

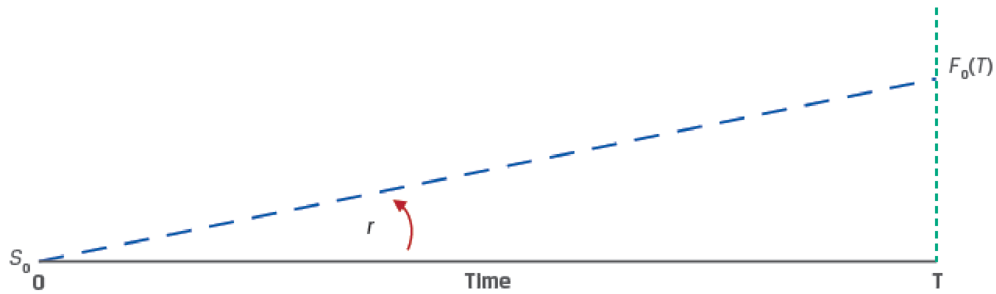
In this example, the spot price of gold is below the PV of the known futures price in three months. Other market participants will also exploit this arbitrage opportunity by purchasing gold in the spot market and selling it in the forward market. This will cause the spot price to increase and the forward price to fall until the arbitrage opportunity is eliminated.

Instructor's Note: The two scenarios demonstrate that arbitrage opportunities will be exploited by market participants and the prices will quickly adjust. Therefore, while pricing derivative we set a price such that no arbitrage opportunities exist.

The two key arbitrage concepts used to price derivatives for an underlying with no additional cash flows are:

- Identical assets or assets with identical cash flows traded at the same time must have the same price
- Assets with a known future price must have a spot price that equals the future price discounted at the risk-free rate

The relationship between spot prices, forward commitment prices, and the risk-free rate is shown in Exhibit 2.



For a given time T, the higher the risk-free rate (r), the higher the forward price.

For a given risk-free rate, the longer the time period (T), the higher the forward price.

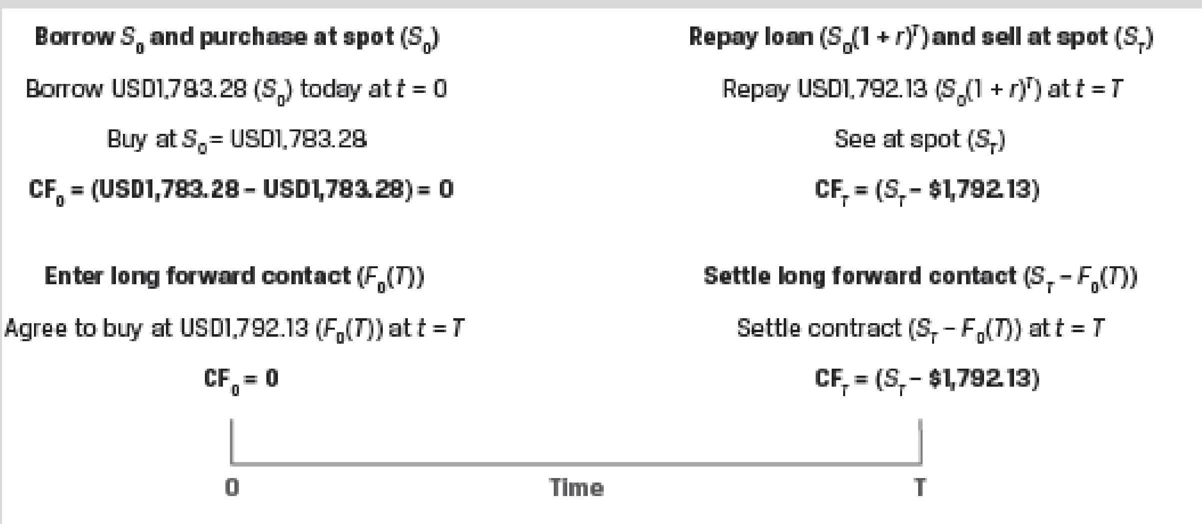
3. Replication

Replication is a strategy in which a derivative's cash flow stream may be recreated using a combination of long or short positions in an underlying asset and borrowing or lending cash.

Example 2 from the curriculum, demonstrates how a forward commitment can be replicated by purchasing the underlying asset in the spot market using funds borrowed at the risk-free rate.

Example: Replication of a Forward Commitment
(*This is based on Example 2 from the curriculum.*)

Using the same data as Example 1, now assume that the spot price of gold has risen to USD1,783.28 so that the arbitrage opportunity is eliminated. Assume again that the risk-free interest rate (r) is 2% and gold can be stored at no cost.



Two equivalent choices available to an investor are:

1. Long forward commitment (agree to buy at $F_0(T)$ at t = T)

- The company enters a forward commitment to buy 100 ounces of gold in three months at a forward price, $F_0(T)$, of USD1,792.13 per ounce.
- If S_T = USD1,900 per ounce, Profit = USD10,787 = 100 × (USD1,900 – USD1,792.13).
- If S_T = USD1,700 per ounce, Profit = –USD(9,213 = 100 × (USD1,700 –USD1,792.13).

2. Borrow and purchase (borrow S_0 and buy asset at S_0 at t = 0)

- The company borrows USD178,328 at 2% and buys 100 ounces of gold at today's spot price, S_0.
- The company sells the gold in three months at spot price S_T.
- The company repays the loan principal and interest: USD179,213 = USD178,328$(1.02)^{0.25}$.
- If S_T = USD1,900, Profit = USD10,787 = 100 × (USD1,900 – USD1,792.13).
- If S_T = USD1,700, Profit = –USD9,213 = 100 × (USD1,700 – USD1,792.13).

When the forward contract is priced such that no arbitrage opportunity exists ($F_0(T) = S_0(1 + r)^T$), both choices will result in the same payoff, i.e., we can say that a forward commitment can be replicated buy purchasing the underlying asset in the spot market using borrowed funds.

Derivatives can also be used to replicate a risk-free rate. When a long cash position is combined with a short forward position, the investor will earn a risk-free rate of return. This is demonstrated in Example 3 from the curriculum.

Example: Risk-Free Trade Replication: Long Asset, Short Forward
(*This is based on Example 3 from the curriculum.*)

A company buys 100 ounces of gold at today's spot price (S_0) of USD1,783.28 and simultaneously enters a forward commitment to sell gold at the forward price, $F_0(T)$, of USD1,792.13.

At t = 0, the company's cash flow is $-S_0$ = –USD178,328.

At t = T, the company's cash flow is $F_0(T)$ = USD179,213.

Solve for the rate of return on this strategy, as follows:

USD179,213 = USD178,328$(1 + r)^{0.25}$.

r = 2.0%, which is equal to the risk-free rate.

4. Costs and Benefits Associated with Owning the Underlying

So far in our discussion we assumed that the underlying asset had no additional associated cash flows and the forward prices were established as:

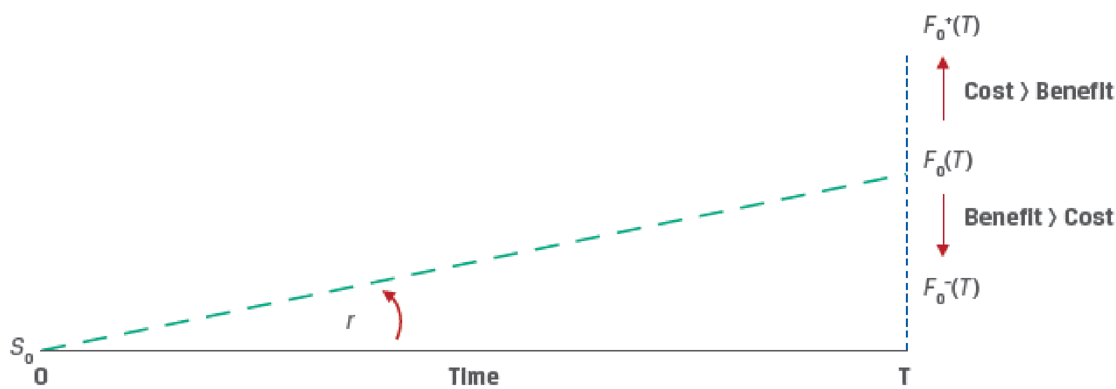
$F_0(T) = S_0(1 + r)^T$ under discrete compounding and

$F_0(T) = S_0e^{rT}$ under continuous compounding

In these formulas, the risk-free rate r denotes the **opportunity cost** of holding the asset. If the investor had not purchased the asset, he could have earned the risk-free rate instead. Therefore, the forward price should be such that the investor is compensated for bearing this cost.

Now, we will relax our assumptions and consider the additional costs and benefits associated with owning underlying assets. Examples of costs of ownership include storage, transportation, insurance and spoilage costs. Examples of benefits of ownership include dividends and coupons.

Refer to Exhibit 7 from the curriculum.



If the underlying asset owner incurs cost in addition to the opportunity costs, the forward price should be higher to compensate investors. On the other hand, if the underlying asset provides benefits the forward price will be lower.

Cost of carry is the net cost of holding an asset, Cost of carry = Cost – Benefits.

The relationship between spot and forward prices for an underlying asset with ownership benefits/income (I) and costs (C) can be expressed as:

$$F_0(T) = [S_0 - PV_0(I) + PV_0(C)](1 + r)^T$$

Sometimes, the additional costs or benefits are expressed as a rate of return instead of absolute amounts. For income (i) and cost (c) expressed as rates of return, the relationship between spot and forward prices under continuous compounding is:

$$F_0(T) = S_0e^{(r+c-i)T}$$

Exhibit 8 illustrates the relationship between forward and spot prices in the presence of costs and benefits.

Cost vs. Benefit	Forward vs. Spot Price
Opportunity and Other Cost > Benefit	$F_0(T) > S_0$
Opportunity and Other Cost < Benefit	$F_0(T) < S_0$
Opportunity and Other Cost = Benefit	$F_0(T) = S_0$

Example: Equity Forward with Dividend
(*This is based on Example 4 from the curriculum.*)

A company enters into a forward contract to deliver 1,000 shares of Unilever (UL) in six months. UL has a spot price of EUR50. It pays a quarterly dividend of EUR0.30, which occurs in exactly three months and again at time T. The risk-free rate is 5%. Calculate the forward price $F_0(T)$, in six months.

Solution:

The forward price can be calculated as:

$F_0(T) = [S_0 - PV_0(I) + PV_0(C)](1 + r)^T$.

As per the data provided, there are no costs in owning UL, therefore $PV_0(C) = 0$

$PV_0(I)$ is the present value of the dividend per share. It can be calculated as:

$PV_0(I) = \text{EUR}0.30(1.05)^{-0.25} + \text{EUR}0.30(1.05)^{-0.5}$.

$PV_0(I) = 0.2964 + 0.2928 = \text{EUR}0.5892$.

Therefore, $F_0(T) = (\text{EUR}50 - \text{EUR}0.5892)(1.05)^{0.5} = \text{EUR}50.6310$.

Example: Stock Index Futures with Dividend Yield
(*This is based on Example 5 from the curriculum.*)

An investor would like to enter into a three-month forward commitment contract to purchase the NIFTY 50 benchmark Indian stock market index. The spot NIFTY 50 index price is INR15,200, the index dividend yield is 2.2%, and the Indian rupee risk-free rate is 4%. Calculate the forward price. (Assume c = 0)

Solution:

The forward price can be calculated as:

$F_0(T) = S_0e^{(r+c-i)T}$

$= 15{,}200e^{(0.04 - 0.022)0.25}$

= INR15,268.55.

FX (foreign exchange) Forward Contracts

An FX (foreign exchange) forward contract involves the sale of one currency and purchase of the other on a future date at a forward price ($F_{0,f/d}$) agreed on at inception.

A long FX forward position involves the purchase of the base currency and the sale of the price currency. For example, a long USD/EUR FX forward position involves purchasing Euros and selling US dollars at a forward rate.

The no-arbitrage condition FX forward rate can be calculated as:

$$F_0(T) = S_0 e^{(r_f - r_d)T}$$

Instructors Note: Foreign exchange is a special case in which the cost of carry is the interest rate differential between the foreign and domestic risk-free rates ($r_f - r_d$).

Exhibit 9 from the curriculum illustrates the relationship between FX Forward and spot rates.

Interest Rate Differential	Forward vs. Spot Price
$(r_f - r_d) > 0$	$F_{0,f/d}(T) > S_{0,f/d}$
$(r_f - r_d) < 0$	$F_{0,f/d}(T) < S_{0,f/d}$
$(r_f - r_d) = 0$	$F_{0,f/d}(T) = S_{0,f/d}$

Example:
(*This is Knowledge check example 4 from the curriculum.*)

An analyst observes that the current spot MXN/USD exchange rate is 19.50, the Mexican peso six-month risk-free rate is 4%, and the six-month US dollar risk free rate is 0.25%. Describe the relationship between the MXN/USD spot and six-month forward rate.

Solution:

The interest rate differential (rf – rd) > 0 calculates as (4% - 0.25% = 3.75%). Therefore, we can expect the forward rate to be higher than the spot rate.

The forward rate can be calculated as:

$$F_0(T) = S_0 e^{(r_f - r_d)T}$$

$F_0(T) = 19.50\ e^{(4\%-0.25\%)0.5} = 19.8691$

Convenience Yield

Convenience yield refers to a non-cash benefit of holding a physical commodity versus a derivative. Sometimes economic conditions may arise that cause market participants to prefer to own physical commodities. For example, if crude oil inventories are very low, refineries may bid up the spot oil price so that forward prices do not fully reflect storage costs and interest rates.

Summary

LO. Explain how the concepts of arbitrage and replication are used in pricing derivatives.

The two key arbitrage concepts used to price derivatives for an underlying with no additional cash flows are:

- Identical assets or assets with identical cash flows traded at the same time must have the same price
- Assets with a known future price must have a spot price that equals the future price discounted at the risk-free rate

Replication is a strategy in which a derivative's cash flow stream may be recreated using a combination of long or short positions in an underlying asset and borrowing or lending cash. For example, a forward commitment can be replicated buy purchasing the underlying asset in the spot market using funds borrowed at the risk-free rate.

When the underlying asset has no additional associated cash flows, the forward prices can be established as:

$F_0(T) = S_0(1 + r)^T$

$F_0(T) = S_0e^{rT}$

LO. Explain the difference between the spot and expected future price of an underlying and the cost of carry associated with holding the underlying asset.

Cost of carry is the net cost of holding an asset, Cost of carry = Cost – Benefits.

The relationship between spot and forward prices for an underlying asset with ownership benefits/income (I) and costs (C) can be expressed as:

$F_0(T) = [S_0 - PV_0(I) + PV_0(C)](1 + r)^T$.

Sometimes, the additional costs or benefits are expressed as a rate of return instead of absolute amounts. For income (i) and cost (c) expressed as rates of return, the relationship between spot and forward prices under continuous compounding is:

$F_0(T) = S_0e^{(r+c-i)T}$

Foreign exchange represents a special case in which the cost of carry is the interest rate differential between two currencies.

$$F_0(T) = S_0e^{(r_f - r_d)T}$$

LM05 Pricing and Valuation of Forward Contracts

Version 1.0

1. Introduction

This learning module covers:

- How the value and price of a forward contract are determined at initiation, during the life of the contract, and at expiration.
- How forward rates are determined from an underlying with a term structure; and their uses.

2. Pricing and Valuation of Forward Commitments

Pricing versus Valuation of Forward Contracts

Let us take an example. Assume Leo owns a share of GE, whose spot price today (S_0) is $100. Rachel enters into a forward contract with Leo today to buy a share of GE at $110 after 1 year.

The terms, parties, and ways to settle the contract are described below:

- $110 is the forward price; the contract is for a period of 1 year.
- Neither Rachel nor Leo pays any money to each other at t = 0 when they enter into the contract.
- Rachel is the long party and Leo is the short party.
- When the contract expires after 1 year, the forward contract can be settled in two ways: 1) Rachel pays $110 to Leo and receives the share in return; 2) if GE trades at $113 after 1 year, then Leo pays $3 (the difference between the agreed-upon forward price and the current stock price) to Rachel.
- There is a risk of default in forward contracts. If the stock price increases to $113, Rachel faces the risk that Leo fails to deliver the share as agreed. Conversely, if the stock price decreases to $99, Leo faces the risk that Rachel does not buy the share as agreed upon.
- Is there an alternative way for Rachel to buy this asset? Yes, to wait until time T and buy the asset at its then spot price S_T. The drawback is that one cannot be sure what the price of the asset will be then; it may be more, or less.

The above example is shown in a generic way in the diagram below. It shows the transactions from a buyer's perspective at times t = 0 when the contract is initiated and time t = T, when the contract expires.

Interpreting the notation and key features of a forward contract:

- *Forward price*: It is denoted by F_0 (T) where subscript 0 indicates the price that is agreed upon at time t = 0 and T indicates the contract ends at time T.
- *Spot price*: It is denoted by S_t. At t = 0, spot price is S_0. When the forward contract expires at t = T, the spot price is S_T. At any time during the life of the contract, it is denoted by S_t.
- *Value of the contract at time t* is denoted by $V_t(T)$.
- The price to be paid at time T when the contract expires is fixed at time t = 0.
- No money exchanges hands at time t = 0, i.e., no party pays anything to the other at t = 0.

The forward price agreed at the initiation date of the contract is the spot rate compounded at the risk-free rate over the life of the contract.

$$F_0(T) = S_0 \text{ x } (1 + r)^T$$

Assuming a risk-free rate of 10%, spot price of $S_0 =$ \$100, and time period of 1 year, we calculate the forward price as:

$$F_0 \text{ (T)} = S_0 \text{ x } (1 + r)^T = 100 \times (1.1)^1 = 110$$

Why is the forward price equal to the spot rate compounded at the risk-free rate over the contract period?

In theory, if Rachel did not enter into the forward contract, then she could invest $100 for period T in a risk-free asset and earn 10% or $110 after 1 year. So, the forward contract must earn the risk-free rate as an excess return will result in an arbitrage opportunity.

Value of a forward contract at initiation

Value of a forward contract at initiation = 0

The value of a forward contract at initiation is zero because neither party pays any money to the other. There is no value to either party.

- At the end of the contract, Leo, as the short party, is obligated to sell the asset to the buyer (Rachel) for $110.
- Given a spot price of $100, a risk-free rate of 10%, and a forward price of $110, there is no advantage to either party at t = 0. The forward price is such that no party can earn an arbitrage profit.

Value at expiration for the long party:

$$V_t(T) = S_T - F_0(T)$$

where:

S_T = spot price of the underlying

$F_0(T)$ = forward price agreed in the contract

The value of a forward contract is positive to the long party if $S_T > F_0(T)$ and negative if $S_T < F_0(T)$ at expiration.

<u>Value at expiration for the short party:</u>

$V_t(T) = -(S_T - F_0(T))$

The value of a forward contract is positive to the short party if $F_0(T) > S_T$ and negative if $F_0(T) < S_T$ at expiration.

If the spot price at time T is 115, then Rachel (long) gets the asset by paying 110 and can immediately sell it for 115 at the then market price. Value to the long = $S_T - F$ = 115 - 110 = 5.

<u>Value of a forward contract during the life of the contract</u>

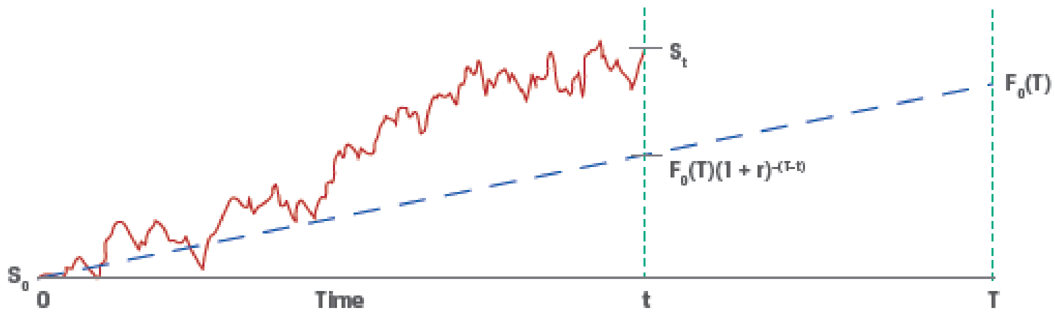
Value during the life of the contract $V_t = S_t - \frac{F}{(1 + r)^{T-t}}$

This is the difference between the spot price at time t, and the present value of forward price for the remaining life of the contract. In our example, after 6 months at t = 0.5, if S_t = 116, then $V_t = 116 - \frac{110}{(1.1)^{(1-0.5)}} = 11.12$. Since the price has gone up, the value to the long party is positive.

Instructor's Note

As market conditions change, only the value of a forward contract changes. Its price does not change as it is fixed at contract initiation.

Thus, the passage of time and spot price changes causes the MTM value of a forward contract to fluctuate representing a gain or loss to contract participants. Exhibit 5 from the curriculum shows this relationship, assuming a constant risk-free rate over the life of a contract for long and short forward contract positions.



Outcome	$V_t(T)$ (long position)	$V_t(T)$ (short position)
$S_t > F_0(T)(1 + r)^{-(T-t)}$	MTM gain	MTM loss
$S_t < F_0(T)(1 + r)^{-(T-t)}$	MTM loss	MTM gain
$S_t = F_0(T)(1 + r)^{-(T-t)}$	No MTM gain or loss	No MTM gain or loss

Pricing and Valuation of Forward Contracts with Additional Costs or Benefits

Now, let us consider an asset with benefits or income(I) and costs (C). How is the forward price for such an asset calculated? The forward price of an asset with benefits and/or costs is the spot price compounded at the risk-free rate over the life of the contract minus the future value of those benefits and costs.

$$F_0\ (T) = \ (S_0 - PV_0(I) + PV_0(C)\ )\ x\ (1\ +\ r)^T$$

where:

$PV_0(I)$ = present value of the benefit. It is subtracted from the spot price because if you own the asset, you receive any benefits associated with the asset, during the life of the contract.

$PV_0(C)$ = present value of costs incurred on the asset during the life of the contract. These costs make it more expensive to hold the asset and hence increase the forward price.

In the previous example, assume $10 is the present value of benefits from the asset and $20 is the present value of costs associated with holding the asset. The forward price at time 0 when Rachel enters into the contract is:

$$F_0\,(T) = \left(100 - 10\ +\ 20\right)(1.1)^1 = 121$$

<u>Value of the forward contract</u>

At any time t over the life of the contract, the MTM value of the forward contract, Vt(T), will depend on the difference between the current spot price adjusted for any remaining costs or benefits from time t through maturity.

$$V_t\ (T) = \left(S_t - PV_t(I) + PV_t(C)\right) - \frac{F_0(T)}{(1+r)^{T-t}}$$

Example: Equity Forward Valuation

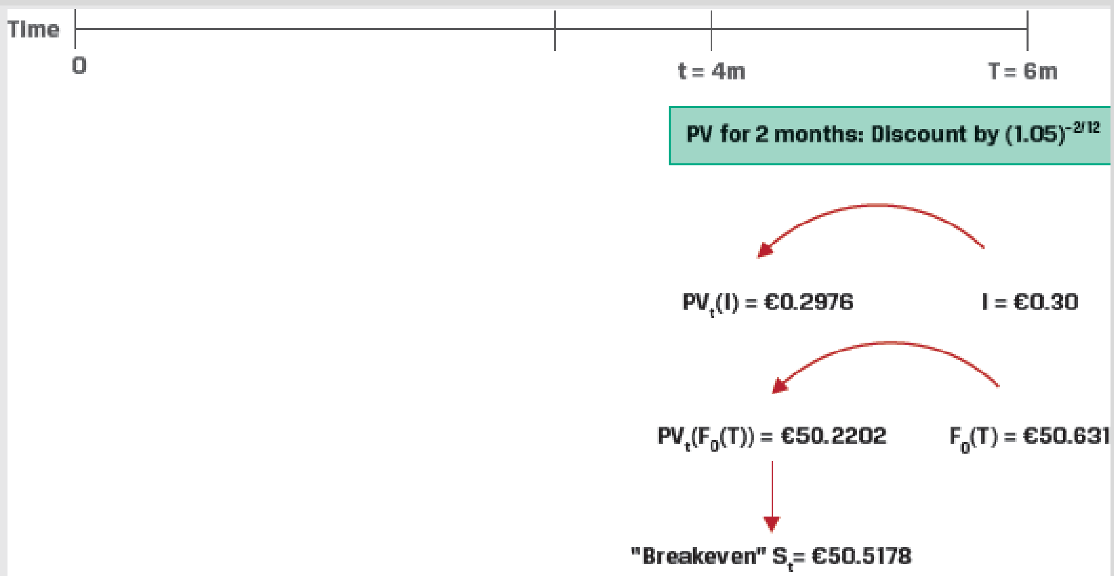
(*This is based on Example 4 from the curriculum.*)

A company agrees to deliver 1,000 Unilever (UL) shares to a financial intermediary in six months under a forward contract at a price of EUR 50.6311 per share. Unilever pays a quarterly dividend of EUR 0.30 three months after contract inception and at time T, and the risk-free rate (r) is 5%. Calculate the forward contract breakeven price, S_t, where $V_t(T)$ = MTM = 0 four months after contract inception if the risk-free rate, r, remains unchanged at 5%.

Solution:

Calculate the present value of the dividend per share, $PV_t(I)$, given that the second dividend will be paid in two months:

$PV_t(I)$ = EUR0.30$(1.05)^{-2/12}$ = EUR0.2976



With PVt(C) = 0, solve for $V_t(T)$ = 0 four months after contract inception,

$$V_t\,(T) = \left(S_t - PV_t(I) + PV_t(C)\right) - \frac{F_0(T)}{(1+r)^{T-t}}$$

$0 = (St - 0.2976) - 50.6311(1.05)^{-2/12}$

St = EUR50.2202 + EUR0.2976

= EUR50.5178

Hightest Capital's breakeven spot rate, S_t (i.e., where $V_t(T) = 0$), four months after inception of the forward contract is therefore EUR50.5178 per share.

Pricing and Valuation of Foreign Exchange Forwards

The forward price for a FX forward contract is calculated as:

$$F_0(T) = S_0 e^{(r_f - r_d)T}$$

At any given time t, the MTM value of the FX forward is the difference between the current spot FX price (S_t) and the present value of the forward price discounted by the current difference in risk-free rates (rf – r_d) for the remaining period through maturity,

$$V_t(T) = S_t - \frac{F_0(T)}{e^{(r_f - r_d)(T-t)}}$$

Example: FX Forward MTM
(*This is based on Example 5 from the curriculum.*)

A company has entered into a long one-year USD/EUR forward contract. It has agreed to purchase EUR1,000,000 in exchange for USD1,201,000 in one year. At time t=0, when the contract is initiated, the USD/EUR spot exchange rate is 1.192 , the one-year USD risk-free rate is 0.50%, and the one-year EUR risk-free rate is −0.25%.

Describe the MTM impact on the FX forward contract from the company's perspective if the one-year USD risk-free rate instantaneously rises by 0.25% once the contract is initiated, with other details unchanged.

Solution:

The MTM value at t=0 can be calculated as:

$$V_t(T) = S_0 - \frac{F_0(T)}{e^{(r_f - r_d)(T)}}$$

S_0 = 1.192

F_0 = 1,201,000/ 1,000,000 = 1.201

r_f = 0.75%

r_d = -0.25%

T = 1

Using these inputs,

$$V_t(T) = 1.192 - \frac{1.201}{e^{(0.0075+0.0025)}} = 0.00295 \text{ USD/EUR}$$

3. Pricing and Valuation of Interest Rate Forward Contracts

Interest Rate Forward Contracts

Yield to Maturity (YTM)

The YTM is the discount rate that, when applied to a bond's promised cash flows, equates those cash flows to the bond's market price.

Suppose we are provided the following data for three recently issued annual fixed-coupon government bonds:

Years to Maturity	Annual Coupon	PV (per 100 FV)
1	1.50%	99.125
2	2.50%	98.275
3	3.25%	98.00

Using this information we can calculate the YTM for each bond as follows:

Bond 1: N = 1, PV = -99.125, PMT = 1.50, FV = 100 →CPT I/Y = 2.3960%

Bond 2: N = 2, PV = -98.275, PMT = 2.50, FV = 100 →CPT I/Y = 3.4068%

Bond 3: N = 3, PV = -98.00, PMT = 3.25, FV = 100 →CPT I/Y = 3.9703%

Zero Rates

A zero rate (also called spot rate) is the YTM on zero-coupon bonds. They are denoted as (z_1, z_2, ...z_N), where z_i is the zero rate for period i.

Zero rates can be computed from the YTM of coupon paying government bonds through a process called 'bootstrapping'.

Continuing with our example, lets add a YTM column to our data on the 3 coupon paying government bonds.

Years to Maturity	Annual Coupon	PV (per 100 FV)	YTM
1	1.50%	99.125	2.3960%
2	2.50%	98.275	3.4068%
3	3.25%	98.00	3.9703%

Starting with the one-year bond, which consists of a single cash flow at maturity, we can solve for the one-year zero rate (z_1) as follows:

One year:

$$99.125 = \frac{101.5}{(1 + z_1)^1}; z_1 = 2.3960\%$$

Instructor's Note: Since the 1-year bond consists of a single cash flow at maturity, the zero rate and the YTM are identical.

Since all cash flows at time t = 1 are discounted at z_1, we can substitute z_1 into the two-year fixed-coupon bond calculation to solve for the zero-coupon rate at the end of the second period (z_2):

Two year:

$$98.275 = \frac{2.5}{1.02396} + \frac{102.5}{(1 + z_2)^2}; z_2 = 3.4197\%$$

We then substitute both z_1 and z_2 into the three-year bond equation to solve for the zero-coupon cash flow at the end of year three (z_3):

Three year:

$$98.00 = \frac{3.25}{1.02396} + \frac{3.25}{1.034197^2} + \frac{103.25}{(1 + z_3)^3}; z_3 = 4.0005\%$$

Discount Factors

The price or present value of a risk-free single-unit payment (e.g., \$1, €1, or £1) after *N* periods is called the discount factor, denoted by DF_N.

Discount factors can be calculated from zero rates using the formula:

$$DF_i = \frac{1}{[1 + Z_i]^i}$$

Continuing with our example, the discount factors for years 1,2, and 3 can be computed as:

One-year: $DF_1 = 1/(1 + z_1)$; $DF_1 = 0.976601$

Two-year: $DF_2 = 1/(1 + z_2)^2$; $DF_2 = 0.934961$

Three-year: $DF_3 = 1/(1 + z_3)^3$; $DF_3 = 0.888982$

A discount factor can also be interpreted as the price of a zero-coupon bond.

Forward Rates

A forward rate is the interest rate that is determined today for a loan that will be initiated at a future time period. $F_{A,B}$ represents a forward rate for a loan initiated A years from today with further maturity of B years. For example, $F_{2,3}$ represents the forward rate for a loan which starts at the end of year 2 and matured at the end of year 5.

Given a set of zero rates we can calculate the 'implied forward rates'. Continuing with our example, let us present the zero rates we calculated in a table.

Years to Maturity	**Zero Rate**
1	2.3960%
2	3.4197%
3	4.0005%

Over a two-year period an investor faces the following investment choices:

1. Invest USD100 for one year today at the zero rate (z_1) of 2.396% and reinvest the

proceeds of USD102.40 at the one-year rate in one year's time, or the implied forward rate ($IFR_{1,1}$).

2. Invest USD100 for two years at the two-year zero rate (z_2) of 3.4197% to receive USD100$(1 + z_2)^2$, or USD106.96.

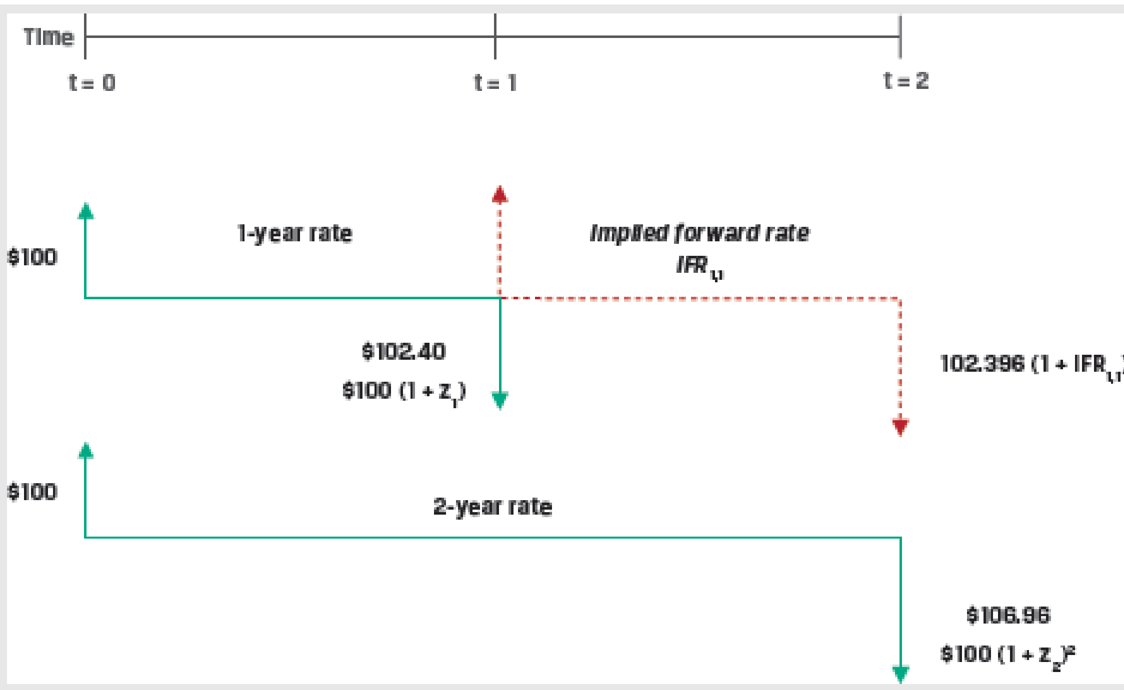
Based on the no-arbitrage principal both options should give the same results. Therefore,

USD100 × $(1 + z_1) \times (1 + IFR_{1,1})$ = USD100 × $(1 + z_2)^2$

USD100 × $(1.02396) \times (1 + IFR_{1,1})$ = USD100 × $(1.034197)^2$

$IFR_{1,1} = 4.4536\%$

This is illustrated in the figure below.



A general formula for the relationship between two spot rates (z_A, z_B) and the implied forward rate ($IFR_{A,B-A}$):

$$(1 + z_A)^A \times (1 + IFR_{A,B-A})^{B-A} = (1 + z_B)^B$$

Using this formula, we can solve for the one-period rate in two periods ($IFR_{2,1}$) as follows:

$(1 + z_2)^2 \times (1 + IFR_{2,1})^1 = (1 + z_3)^3$

$(1.034197)^2 \times (1 + IFR_{2,1}) = (1.040005)^3$

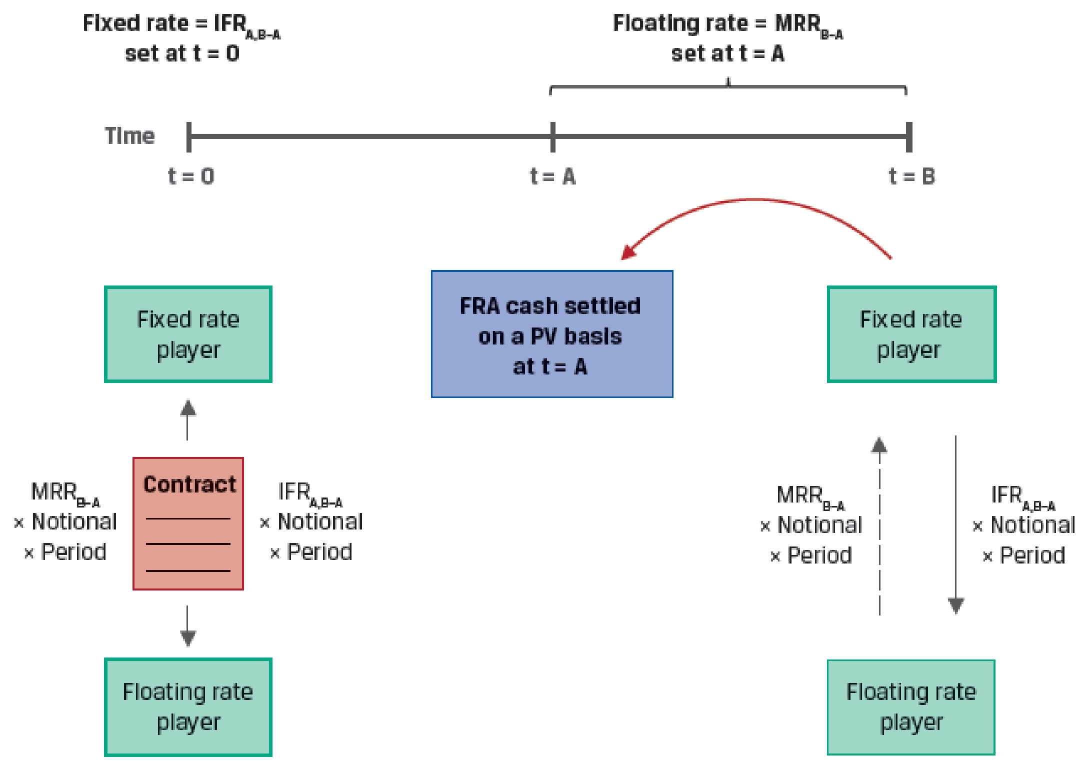
$IFR_{2,1} = 5.1719\%$

Forward Rate Agreements (FRAs)

A forward rate agreement (FRA) is an OTC derivatives contract in which counterparties agree to apply a specific interest rate to a future period. FRAs allow us to lock in a rate today for a loan in the future and act as a hedge against interest rate risk.

The FRA buyer, or long position, agrees to pay an interest based on a fixed rate and receive interest based on a future market reference rate. The interest payments are calculated on a hypothetical notional amount.

Exhibit 9 from the curriculum shows the FRA mechanics at t = 0 and settlement at time A.



The payoffs on an FRA are determined by market interest rates at expiration:

- If reference rate at FRA expiration is greater than the FRA rate, then the long benefits.
- If reference rate at FRA expiration is lower than the FRA rate, then the short benefits.

Example: FRA

(This is Example 5 from Knowledge check)

A counterparty agrees to be the FRA fixed-rate receiver on a one-month AUD MRR in three months' time based on a AUD150,000,000 notional amount. If $IFR_{3m,1m}$ at contract inception is 0.50% and one-month AUD MRR sets at 0.35% for settlement of the contract, calculate the settlement amount and interpret the results.

Solution:

A short FRA position, or fixed-rate receiver, agrees to pay interest based on a market reference rate determined at settlement and receives interest based on a pre-agreed fixed rate, the implied forward rate ($IFR_{3m,1m}$). Since AUD MRR_{B-A} at settlement is below the pre-agreed fixed rate, the FRA fixed-rate receiver realizes a gain and receives a net payment based on the following calculation:

Net Payment = ($IFR_{A,B-A}$ – MRR_{B-A}) × Notional Principal × Period

= (0.50% – 0.35%) × AUD150,000,000 × (1/12)

= AUD18,750 at the end of the period

Solve for the present value of settlement given the contract is settled at the beginning of the period, using MRR_{B-A} as the discount rate:

Cash Settlement (PV) = AUD18,750.00/(1 + 0.35%/12)

= AUD18,744.53

Summary

LO. Explain how the value and price of a forward contract are determined at initiation, during the life of the contract, and at expiration.

Forward Contracts with no Additional Costs and Benefits:

The forward price agreed at the initiation date of the contract is the spot rate compounded at the risk-free rate over the life of the contract.

$$F_0(T) = S_0 \text{ x } (1 + r)^T$$

The price remains fixed, however the MTM value of the forward contract changes based on the passage of time and the underlying spot price.

- At initiation the value of forward contract is zero.
- Value at expiration: S_T - F
- Value during the life of the contract: $V_t = S_t - \frac{F}{(1+r)^{T-t}}$

Forward Contracts with Additional Costs and Benefits:

The forward price is calculated as:

$$F_0\ (T) = (S_0 - PV_0(I) + PV_0(C)\ ) \text{ x } (1 + r)^T$$

At any time t over the life of the contract, the MTM value of the forward contract is calculated as:

$$V_t\ (T) = \left(S_t - PV_t(I) + PV_t(C)\right) - \frac{F_0(T)}{(1+r)^{T-t}}$$

Pricing and Valuation of FX Forwards:

The forward price for a FX forward contract is calculated as:

$$F_0(T) = S_0 e^{(r_f - r_d)T}$$

At any given time t, the MTM value of the FX forward is calculated as:

$$V_t(T) = S_t - \frac{F_0(T)}{e^{(r_f - r_d)(T-t)}}$$

LO. Explain how forward rates are determined for an underlying with a term structure and describe their uses.

Yield to Maturity: The YTM is the discount rate that, when applied to a bond's promised cash flows, equates those cash flows to the bond's market price.

Zero Rates: A zero rate (also called spot rate) is the YTM on zero-coupon bonds. Zero rates can be computed from the YTM of coupon paying government bonds through a process called 'bootstrapping'.

Discount Factors: The price or present value of a risk-free single-unit payment (e.g., $1, €1, or £1) after N periods is called the discount factor, denoted by DF_N. Discount factors can be calculated from zero rates using the formula:

$$\mathrm{DF}_i = \frac{1}{[1 + \mathrm{Z}_i]^i}$$

Forward Rates: A forward rate is the interest rate that is determined today for a loan that will be initiated at a future time period. $F_{A,B}$ represents a forward rate for a loan initiated A years from today with further maturity of B years.

A general formula for the relationship between two spot rates (z_A, z_B) and the implied forward rate ($IFR_{A,B-A}$):

$$(1 + z_A)^A \times (1 + IFR_{A,B-A})^{B-A} = (1 + z_B)^B$$

Forward Rate Agreements (FRAs): A FRA is an OTC derivatives contract in which counterparties agree to apply a specific interest rate to a future period. The FRA buyer, or long position, agrees to pay an interest based on a fixed rate and receive interest based on a future market reference rate. The interest payments are calculated on a hypothetical notional amount.

The payoffs on an FRA are determined by market interest rates at expiration:

- If reference rate at FRA expiration is greater than the FRA rate, then the long benefits.
- If reference rate at FRA expiration is lower than the FRA rate, then the short benefits.

LM06 Pricing and Valuation of Futures Contracts

Version 1.0

1. Introduction

This learning module covers:

- Pricing and valuation of futures contracts and how it differs from forward contracts
- Interest rate futures
- Effect of central clearing of OTC derivatives

2. Pricing of Futures Contracts at Inception

Futures are standardized, exchange-traded derivatives (ETDs) with zero initial value and a futures price $f_0(T)$ established at inception.

Instructor's Note: Small letter 'f' is used for futures and capital letter 'F' is used for forwards. The calculation of futures price is similar to the calculation of forward price covered in the previous reading.

The futures price for an underlying asset with no cost or benefit is calculated as:

$$f_0(T) = S_0(1 + r)^T$$

As with forwards, discrete compounding is used for futures on individual assets. However, if the underlying asset comprises a portfolio-such as equity, fixed income, commodity or credit index; or if the underlying involves foreign exchange, continuous compounding is used. The futures price is calculated as:

$$f_0(T) = S_0e^{rT}$$

The futures price for an underlying asset with ownership benefits or income (I) and costs (C) is calculated as:

$$f_0(T) = [S_0 – PV_0(I) + PV_0(C)]\ (1 + r)^T$$

3. MTM Valuation: Forwards versus Futures

<u>Marking to market</u>: Futures contracts are marked to market on a daily basis. What this means is that if there is a gain or loss on the position relative to the previous day's closing price, then the gain is credited to the winning account by deducting the amount from the losing account.

So unlike a forward contract, where the gain or loss is realized at the end of the contract period; in a futures contract, there is a cash flow on a daily basis. This daily settlement resets the futures contract value to zero every day and the process continues until contract maturity when the futures price converges to the spot price S_T.

The cumulative realized mark-to-market (MTM) gain or loss on a futures contract is approximately the same as for a comparable forward contract. However, while the forward contract price, $F_0(T)$ remains fixed until the contract matures, future contract prices fluctuate daily based on market changes.

The following example will help understand these concepts.

Example:

Assume that the spot price of a stock is $100, and that a three-period futures contract on this stock has a price of $115. An equivalent forward contract expiring in three periods also has a price of $115. The following table shows the cash flows to long and short parties of these contracts as the price of the stock's future contract changes:

Time period	Future contract price	Forward contract price	Long Forward Cash Flow	Short Forward Cash Flow	Long Future Cash Flow	Short Future Cash Flow
0	$115	$115	-	-	-	-
1	$120	$115	$0	$0	$5	-$5
2	$130	$115	$0	$0	$10	-$10
3	$125	$115	$10	-$10	-$5	$5
Net			$10	-$10	$10	-$10

In both cases, the net cash flow from seller to buyer is $10, but the timing of the cash flows is different. The settlement of a forward contract occurs at maturity. But the profits or losses are recorded on the futures contract at the end of each period.

Also, the forward price remains fixed at $115, but the futures price varies each period.

4. Interest Rate Futures versus Forward Contracts

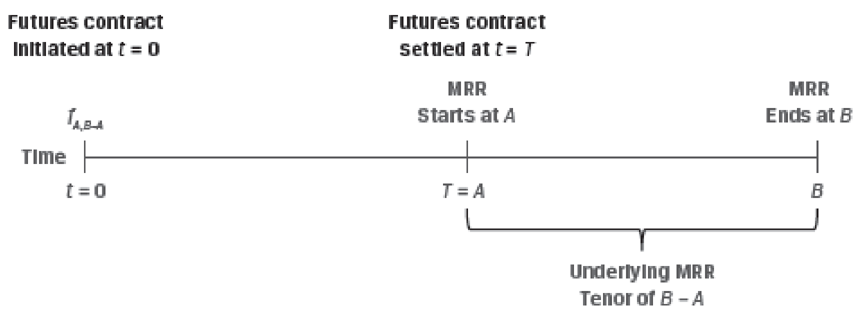
In the previous reading, we learnt that zero rates can be used to calculate the implied forward rate, and that the implied forward rate is used to set the fixed rate on a forward rate agreement (FRA).

Interest rate futures offer investors a highly liquid and standardized alternative to FRA. Like FRAs, the underlying for interest rate futures is the market reference rate (MRR) on a hypothetical deposit. However, interest rate futures trade on a price basis, as per the following formula:

$$f_{A,B-A} = 100 - (100 \times MRR_{A,B-A})$$

where $f_{A,B-A}$ represents the futures price for a market reference rate for B−A periods that begins in A periods ($MRR_{A,B-A}$)

This is illustrated in Exhibit 3 from the curriculum.



The interest rate futures price can be used to calculate the implied MRR. For example, we may solve for the implied three-month MRR rate in three months' time (where A = 3m, B = 6m, B − A = 3m) if an interest rate futures contract is trading at a price of 98.25.

$f_{3m,3m}$: 98.25 = 100 − (100 × $MRR_{A,B-A}$)

$MRR_{3m,3m}$ = 1.75%.

This (100-yield) price convention results in an inverse price/yield relationship that is different from an FRA. A long futures contract gains as price rises i.e. the future MRR falls. Whereas, a short futures contract gains as the price falls i.e. future MRR rises.

Exhibit 4 from the curriculum summarizes the relationship between interest rate futures and FRAs.

Contract Type	Gains from Rising MRR	Gains from Falling MRR
Interest rate futures	Short futures contract	Long futures contract
FRA	Long FRA: FRA fixed-rate payer	Short FRA: FRA floating-rate payer

The daily settlement of interest rate futures is based on price changes, which translate into futures contract basis point value (BPV) as follows:

Futures Contract BPV = Notional Principal × 0.01% × Period

For example, assuming a $1,000,000 notional for three-month MRR of 2.21% for one quarter (or 90/360 days), the underlying deposit contract value would be:

$1,005,525 = $1,000,000 × [1 + (2.21% / 4)].

Consider how a one basis point (0.01%) change in MRR affects contract value:

1 bp increase (2.22%): $1,005,550 = $1,000,000 × [1 + (2.22% / 4)].

1 bp decrease (2.20%): $1,005,500 = $1,000,000 × [1 + (2.20% / 4)].

Both the increase and decrease in MRR by one basis point change the contract BPV by $25. Thus, short-term interest rate futures exhibit a fixed linear relationship between price and yield changes.

5. Forward and Futures Price Differences

Forward and future prices are identical if:

- Interest rates are constant, or
- Future prices and interest rates are uncorrelated

A violation of these assumptions can give rise to differences in pricing between these two contracts.

- If futures prices are positively correlated with interest rates, long futures contracts are more desirable than long forward contracts. This is because rising prices will lead to profits that can be reinvested in periods of rising interest rate, and falling prices lead to losses that occur in periods of falling interest rates.
- Conversely, if future prices are negatively correlated with interest rates, then long forward positions are more desirable than long future positions.
- The more desirable contract will tend to have the higher price. The price difference depends on the volatility of the interest rates.

6. Interest Rate Forward and Futures Price Differences

There is a price difference between interest rate futures and FRAs due to the convexity bias. This is illustrated in the following example.

Example: Interest Rate Forwards versus Futures
(*This is Example 5 from the curriculum.*)

Let us return to an earlier example with an interest rate futures contract of $1,000,000 notional for three-month MRR of 2.21% for one quarter (or 90/360 days). Recall that the underlying deposit contract value was:

$1,005,525 = $1,000,000 × [1 + (2.21% / 4)].

The contract BPV was shown to be $25 (= $1,000,000 × 0.01% × [1/4]).

Consider in contrast a $1,000,000 notional FRA on three-month MRR in three months' time with an identical 2.21% rate. The net payment on the FRA is based upon the difference between MRR and the implied forward rate (IFR):

Net Payment = (MRR_{B-A} − $IFR_{A,B-A}$) × Notional Principal × Period.

For example, if the observed MRR in three months is 2.22% (+0.01%), the net payment at maturity would be $25 (= $1,000,000 × 0.01% × [1/4]). However, the settlement of an FRA is based upon the present value of the final cash flow discounted at MRR, so:

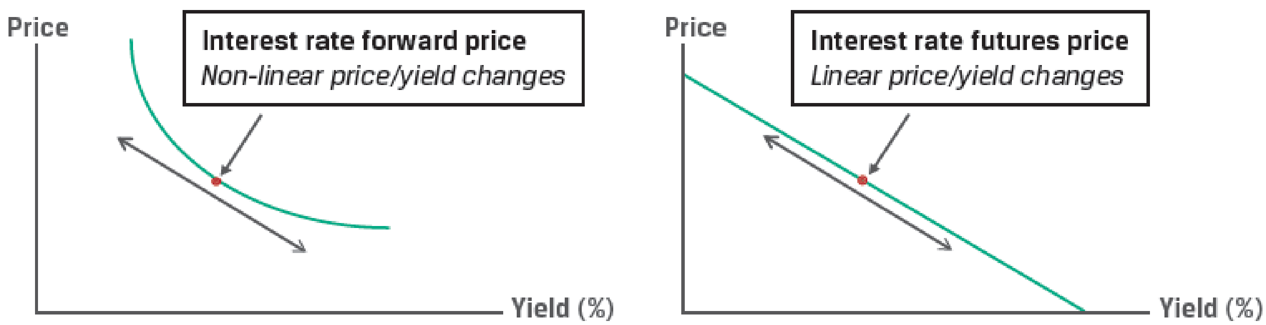
Cash Settlement (PV): $24.86 = $25 /(1 + 0.0222 / 4).

If we increase the magnitude of the MRR change at settlement and compare these changes between a long interest rate futures position and a short receive-fixed (pay floating) FRA contract, we arrive at the following result:

$MRR_{3m,3m}$	**Short FRA Cash Settlement (PV)**	**Long Futures Settlement**
2.01%	$497.50	$500
2.11%	$248.69	$250
2.21%	$0	$0
2.31%	($248.56)	($250)
2.41%	($497.01)	($500)

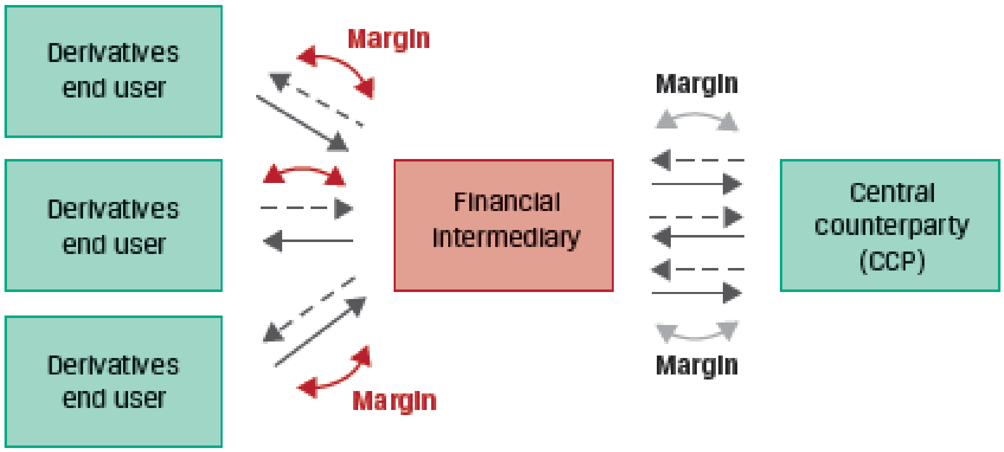
Note that while the futures contract has a fixed linear payoff profile for a given basis point change, the FRA settlement does not.

The discounting feature of the FRA, which is not present in the futures contract, leads to a convexity bias, as illustrated in Exhibit 5.



7. Effect of Central Clearing of OTC Derivatives

The advent of derivatives central clearing has introduced futures-like margining requirements for OTC derivatives dealers who buy and sell forwards to derivatives end users. Dealers who must post cash or highly liquid securities to a central counterparty often impose similar requirements on derivatives end users. This is illustrated in Exhibit 6 from the curriculum.



These margin requirements have reduced the difference in the cash flow impact of ETDs and OTC derivatives. Therefore, the price difference between futures and forwards has also reduced.

Summary

LO. Compare the value and price of forward and futures contracts.

Futures are standardized, exchange-traded derivatives (ETDs) with zero initial value and a futures price $f_0(T)$ established at inception. The futures price for an underlying asset with no cost or benefit is calculated as:

$$f_0(T) = S_0(1 + r)^T$$

Unlike a forward contract, where the gain or loss is realized at the end of the contract period; in a futures contract, there is a cash flow on a daily basis. This daily settlement resets the futures contract value to zero every day and the process continues until contract maturity when the futures price converges to the spot price S_T.

The cumulative realized mark-to-market (MTM) gain or loss on a futures contract is approximately the same as for a comparable forward contract. However, while the forward contract price, $F_0(T)$ remains fixed until the contract matures, forward contract prices fluctuate daily based on market changes.

LO. Explain why forward and futures prices differ.

Forward and future prices are identical if:

- Interest rates are constant, or
- Future prices and interest rates are uncorrelated

A violation of these assumptions can give rise to differences in pricing between these two contracts.

- If futures prices are positively correlated with interest rates, long futures contracts are more desirable than long forward contracts.
- Conversely, if future prices are negatively correlated with interest rates, then long forward positions are more desirable than long future positions.
- The more desirable contract tends to have a higher price. The price difference depends on the interest rate volatility.

<u>Interest rate forward and future price differences</u>: The discounting feature of the FRA, which is not present in the futures contract, leads to a convexity bias. Because of this convexity bias, there is a price difference between interest rate futures and FRAs.

LM07 Pricing and Valuation of Interest Rates and Other Swaps

Version 1.0

1. Introduction

This learning module covers:

- How swap contracts are similar to but different from a series of forward contracts
- Valuation and pricing of swaps

2. Swaps vs. Forwards

A swap contract is an agreement between two counterparties to exchange a *series* of future cash flows, while a forward contact is an agreement for a *single* exchange of value at a future date.

How do Swaps work?

To understand how swaps work, let us consider a 3-year plain vanilla interest rate swap with annual settlement where the fixed-rate payer pays 10% and the floating rate payer pays an interest based on MRR. Also assume that the future MRR rates at t = 0, 1, and 2 are 9%, 10%, and 11%.

Net gain to Fixed	-1%	0%	1%
Fixed payment	10%	10%	10%
Floating payment	9%	10%	11%

At times 0, 1, and 2, the two parties exchange a series of payments. The fixed rate payer makes a fixed payment of 10% and receives a floating payment based on the value of the underlying at that point in time. Here is it 9%, 10%, and 11% respectively.

The payments are netted. At t=1, the fixed rate payer will make a payment of 1% of notional principal to the floating rate payer. At t=2 no cash flows will be exchanged. At t=3, the floating rate payer will make a payment of 1% of notional principal to the fixed rate payer.

Similarities between FRAs and Swaps

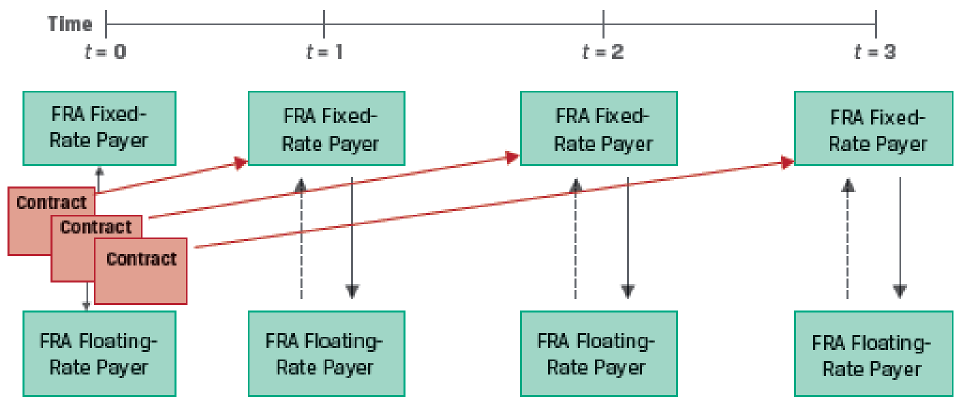
- Both FRAs and swaps allow users to lock-in a fixed rate for future. In both cases, the net difference between a fixed rate agreed upon at inception and a future MRR is used to determine cash settlement.
- Both FRAs and swaps have a symmetric payoff profile.
- No cash is exchanged upfront. Both have a value of 0 at initiation.
- Both FRAs and swaps involve counterparty credit exposure.

Differences between FRAs and Swaps

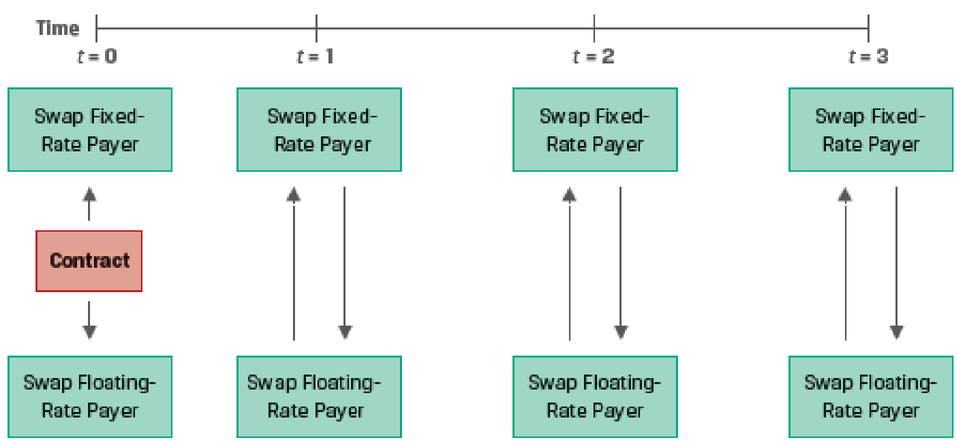
- An FRA has a single settlement, which occurs at the *beginning* of an interest period, while a swap has periodic settlements, which occur at the *end* of each respective period.

- A swap contract is like a series of FRAs, where each FRA has a different time to maturity. While a swap has a *constant* fixed rate over its life, the series of FRAs will have *different* fixed rates for different times to maturity. This is illustrated in Exhibit 2 from the curriculum.

Series of Forward Rate Agreements (at Different Fixed Rates)



Standard Interest Rate Swap (At A Constant Fixed Rate)



- Financial intermediaries often use FRAs to manage interest rate exposure. But derivative end users, such as issuers and investors, usually prefer swaps because they better match rate-sensitive assets and liabilities with periodic cash flows, such as fixed-coupon bonds, variable rate loans, or known future commitments.

Calculating the Swap Rate

The following example demonstrates how to calculate the swap rate.

Example: Swaps as a Combination of Forwards
(*This is based on Example 1 from the curriculum.*)

Suppose we are provided the following data:

Years to maturity	Zero Rates
1	2.3960%
2	3.4197%
3	4.0005%

As shown in a previous reading, using the zero rates we can calculate the implied one period forward rates.

$IFR_{0,1}$ = (1.023960) – 1 =2.3960%

$IFR_{1,1}$ = $(1.034197)^2/(1.02396)$ – 1 = 4.4536%

$IFR_{2,1}$ = $(1.040005)^3/(1.034197)^2$ – 1 = 5.1719%

The par swap rate is the fixed rate that equates the present value of all future expected floating cash flows to the present value of fixed cash flows.

Σ PV(Floating payments) = Σ PV(Fixed payments), or

$$\sum_{i=1}^{N} \frac{\mathrm{IFR}}{(1+z_i)^i} = \sum_{i=1}^{N} \frac{s_i}{(1+z_i)^i}$$

In our example,

$$\frac{\mathrm{IFR}_{0,1}}{(1+z_1)^1} + \frac{\mathrm{IFR}_{1,1}}{(1+z_2)^2} + \frac{\mathrm{IFR}_{2,1}}{(1+z_3)^3} = \frac{s_3}{(1+z_1)^1} + \frac{s_3}{(1+z_2)^2} + \frac{s_3}{(1+z_3)^3}$$

$$\frac{2.3960\%}{(1.023960)^1} + \frac{4.4536\%}{(1.034197)^2} + \frac{5.1719\%}{(1.04005)^3} = \frac{s_3}{(1.02396)^1} + \frac{s_3}{(1.034197)^2} + \frac{s_3}{(1.04005)^3}$$

0.111017 = 2.800545 × s_3.

s_3 = 3.9641%.

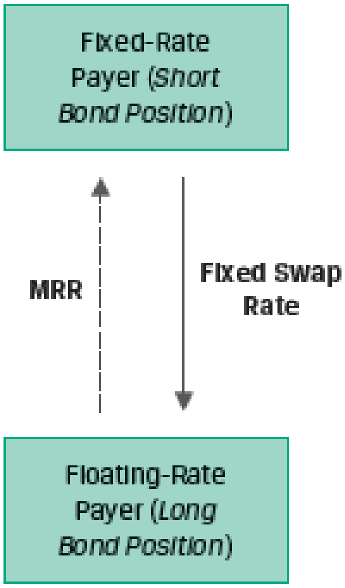
The three-year swap rate of 3.9641% calculated in the above example can be interpreted as a multiperiod breakeven rate at which an investor will be indifferent to:

- paying the fixed swap rate and receiving the respective forward rates or
- receiving the fixed swap rate and paying the respective forward rates.

Therefore, the fixed swap rate can be thought of as an internal rate of return on the implied forward rates over the life of the swap.

Using Swaps to Manage Fixed-Income Exposure

A pay fixed receive floating swap is similar to going long a floating rate bond and going short a fixed rate bond. This is illustrated in Exhibit 4 from the curriculum.



Because swaps are more liquid than individual bond positions, active fixed-income portfolio managers often use swaps rather than underlying securities to adjust their interest rate exposure.

For example, if interest rates are expected to fall, a portfolio manager may choose to receive fixed on a swap rather than purchase underlying bonds in order to profit from the lower rate environment.

Instructor's Note: A receive fixed swap will benefit when interest rates decline. A pay fixed swap will benefit when interest rates rise.

3. Swap Values and Prices

The swap price is the swap fixed rate determined at contract initiation.

The value of the swap is zero at initiation, but its value changes as time passes and interest rates change.

The periodic settlement for a fixed-rate payer on a swap can be calculated as:

Periodic settlement value = (MRR – swap fixed rate) × Notional amount × Period

The value of a swap on any settlement date equals the current settlement value plus the present value of all remaining future swap settlements.

If the MRR at a settlement date is higher than the expected forward rate (IFR), the PV of floating rate payments will be higher than the PV of fixed rate payments, causing an MTM gain for the fixed-rate payer and an MTM loss for the fixed-rate receiver.

Summary

LO. Describe how swap contracts are similar to but different from a series of forward contracts.

<u>Similarities between FRAs and Swaps</u>

- Both FRAs and swaps allow users to lock-in a fixed rate for future. In both cases, the net difference between a fixed rate agreed upon at inception and a future MRR is used to determine cash settlement.
- Both FRAs and swaps have a symmetric payoff profile.
- No cash is exchanged upfront. Both have a value of 0 at initiation.
- Both FRAs and swaps involve counterparty credit exposure.

<u>Differences between FRAs and Swaps</u>

- An FRA has a single settlement, which occurs at the *beginning* of an interest period, while a swap has periodic settlements, which occur at the *end* of each respective period.
- A swap contract is like a series of FRAs, where each FRA has a different time to maturity. While a swap has a *constant* fixed rate over its life, the series of FRAs will have *different* fixed rates for different times to maturity.

LO. Contrast the value and price of swaps.

A swap is priced by solving for the fixed rate that equates the present value of all future expected floating cash flows to the present value of fixed cash flows. It can also be thought of as an internal rate of return on the implied forward rates over the life of the swap.

The value of the swap is zero at initiation, but its value changes as time passes and interest rates change. The value of a swap on any settlement date equals the current settlement value plus the present value of all remaining future swap settlements. A rise in the expected forward rates will result in a MTM gain for the fixed rate payer (floating rate receiver).

LM08 Pricing and Valuation of Options

Version 1.0

1. Introduction

This learning module covers:

- The exercise value, moneyness, and time value of an option
- How arbitrage and replication concepts can be applied to options
- Factors affecting the value of an option

2. Option Value relative to the Underlying Spot Price

Options have a non-linear or asymmetric payoff. Their value depends on whether the spot price crosses an exercise threshold at maturity. To gauge an option's value before maturity, investors frequently rely on three measures – exercise value, moneyness, and time value.

Recall from an earlier reading that American options can be exercised at any time, while European options can be exercised only at maturity. This reading will only focus on European options.

3. Option Exercise Value

At expiration, the value of a call and put option can be calculated as:

$$C_T = \max(0, S_T - X)$$

$$P_T = \max(0, X - S_T)$$

At any time before expiration ($t < T$), an option's value is made up of its exercise value and its time value. The exercise value is the option's value if it could be exercised immediately, whereas the time value captures the possibility that the passage of time and the volatility of the underlying price will increase the profitability of exercise at maturity.

<u>Calculating an Option's Exercise Value</u>: At any time before expiration ($t < T$), we can calculate an option's exercise value by comparing the underlying spot price (S_t) with the present value of the exercise price (X).

$$\text{Call Option Exercise Value} = \max\left(0, S_t - \frac{X}{(1+r)^{(T-t)}}\right)$$

$$\text{Put Option Exercise Value} = \max\left(0, \frac{X}{(1+r)^{(T-t)}} - S_t\right)$$

Example: Put Option Exercise Value

(*This is based on Example 1 from the curriculum.*)

Consider a 1-year put option with an exercise price of EUR 1,000, an initial price of EUR 990, and a risk-free rate of 1%. What is the exercise value of the option in six months if the spot price equals EUR 950?

Solution:

$$\text{Put Option Exercise Value} = \max\left(0, \frac{X}{(1+r)^{(T-t)}} - S_t\right)$$

$$\text{Put Option Exercise Value} = \max\left(0, \frac{1{,}000}{(1+0.01)^{(0.5)}} - 950\right) = \max(0, 45.04) = 45.04$$

Similarity with Forwards:

If we assume an exercise price X, equal to the forward price $F_0(T)$, the exercise value of a call option is the same as the value of a long forward commitment: $S_t - PV(F_0(T))$, provided that the spot price is greater than the present value of X.

Note that this comparison ignores the upfront call premium paid by the option buyer.

4. Option Moneyness

An option's moneyness expresses the relationship between the underlying price and the exercise price. It refers to whether an option would result in a profit or loss if exercised immediately.

- If immediate exercise of the option would result in a positive payoff, then the option is in the money.
- If immediate exercise would result in loss, then the option is out of the money.
- If immediate exercise would result in neither a gain nor a loss, then the option is at the money.

The table below shows the moneyness of an option for various relative values of S and X.

	Call Option	**Put Option**
In the money	S > X	S < X
At the money	S = X	S = X
Out of the money	S < X	S > X

Moneyness is used to compare options on the same underlying but with different exercise prices and/or times to maturity.

Also, the degree to which an option is in or out of the money affects the sensitivity of its price to changes in the price of the underlying.

- Deep-in-the-money options, are highly likely to be exercised, and are therefore highly sensitive to changes in the price of the underlying.
- Whereas, deep-out-of-the money options, are very unlikely to be exercised, and are therefore far less sensitive to changes in the price of the underlying.

Example: Put Option Moneyness
(*This is Example 2 from the curriculum.*)

Recall in Example 1 that at time t, a put option with six months remaining to maturity had an exercise price (X) of EUR 1,000 and an underlying spot price (S_t) of EUR 950. Describe the moneyness of the put option at time t.

Solution:

Given that the underlying spot price is below the exercise price, the put option is in the money by EUR 50.

5. Option Time Value

Prior to expiration an option also has time value in addition to exercise value. The time value of an option is the amount by which the option premium exceeds the exercise value.

The longer the time to expiration the higher the time value.

The time value of an option is always positive but it declines to zero at maturity. This process is referred to as **time value decay**.

Example: Put Option Time Value
(*This is based on Example 3 from the curriculum.*)

Example 1 showed that a one-year put option with an exercise price (X) of EUR 1,000 had an exercise value of EUR 45.04 with six months remaining to maturity when the spot price (S_t) was EUR 950. If we observe a current put option price (p_t) of EUR 50, what is the time value of the put option?

Solution:

Put option price = Exercise value + Time value

50 = 45.04 + Time value

Time value = 4.96

6. Arbitrage

A key distinction between forwards and options is that a forward buyer enters into a contract with no cash paid upfront and has an unlimited gain or loss; in contrast, an option buyer pays an upfront premium and will exercise an option at maturity only if it is in the money. This conditional nature of payoff establishes an upper and lower no-arbitrage price bounds for an option at any time t.

Upper and Lower Bounds for Call Options

The lower bound of a call option is its exercise value. An option which trades below its exercise value violates the no-arbitrage condition.

Similarly, the upper bound of a call option is the price of the underlying. A call buyer will not pay more for the right to purchase an underlying than the price of the underlying.

Therefore,

Call option lower bound $= \max\left(0, S_t - \frac{X}{(1+r)^{(T-t)}}\right)$

Call option upper bound $= S_t$

$$\max\left(0, S_t - \frac{X}{(1+r)^{(T-t)}}\right) < c_t \leq S_t$$

Upper and Lower Bounds for Put Options

The lower bound of a put option is its exercise value. An option which trades below its exercise value violates the no-arbitrage condition.

The upper bound of a put option is the exercise price, X. A put option buyer will exercise only if the spot price is below X at maturity.

Therefore,

Put Option Lower Bound $= \max\left(0, \frac{X}{(1+r)^{(T-t)}} - S_t\right)$

Put Option Upper Bound $= X$

$$\max\left(0, \frac{X}{(1+r)^{(T-t)}} - S_t\right) < p_t \leq X$$

Example: Call Option Upper and Lower Bounds
(*This is Example 4 from the curriculum.*)

Consider a one-year call option with an exercise price, X, of EUR 1,000. The underlying asset, S_0, trades at EUR 990 at time t = 0 and the risk-free rate, r, is 1%. What are the no-arbitrage upper and lower bounds in six months' time if the underlying asset price, S_t, equals EUR 1,050?

Solution:

Call option lower bound $= \max\left(0, S_t - \frac{X}{(1+r)^{(T-t)}}\right)$

Call option lower bound $= \max\left(0, 1{,}050 - \frac{1{,}000}{(1+0.01)^{(0.5)}}\right) = \max(0, 54.96) = 54.96$

Call option upper bound $= S_t$

Call option upper bound $= 1{,}050$

7. Replication

Call Option Replication

To replicate a call option at contract inception (t=0), we must borrow at the risk-free rate, r, and use the proceeds to purchase the underlying at a price of S_0.

At expiration, there are two possible replication outcomes:

- Exercise ($S_T > X$): We should sell the underlying for S_T and use the proceeds to repay X.
- No exercise ($S_T < X$): No settlement is required.

If exercise were certain, we would borrow $\frac{X}{(1+r)^T}$ at inception(as is the case of a forward).

However, since exercise is uncertain, we borrow a proportion of $\frac{X}{(1+r)^T}$ based on the likelihood of exercise. This proportion depends on the moneyness of the option and the replicating transaction has to be adjusted as the likelihood of exercise changes. Therefore, as compared to replicating trades of a forward which remain constant, the replicating trades of an option are dynamic.

Put Option Replication

To replicate a put option at contract inception (t=0), we must sell the underlying short at a price of S_0 and lend the proceeds at the risk-free rate, r.

At expiration, there are two possible replication outcomes:

- Exercise ($S_T < X$): We should purchase the underlying for S_T from the proceeds of the risk free loan.
- No exercise ($S_T > X$): No settlement is required.

If exercise were certain, we would lend $\frac{X}{(1+r)^T}$ at inception(as is the case of a forward).

However, as is the case with call option, since exercise is uncertain, we lend a proportion of $\frac{X}{(1+r)^T}$ based on the likelihood of exercise. The replicating transaction has to be adjusted as the likelihood of exercise changes.

8. Factors Affecting Option Value

Several factors affect the value of call and put options. We list these factors below:

Value of the Underlying

A call option becomes more valuable as the spot price of the underlying rises. Whereas, A put option becomes more valuable as the spot price of the underlying falls.

Relationship between the value of the option and the value of the underlying

- The value of a call option is directly related to the value of the underlying.
- The value of a put option is inversely related to the value of the underlying.
- The more 'in-the-money' an option is, the more sensitive the option's value to the underlying price.

Exercise Price

Consider two call options with similar attributes (time to expiration, underlying) but different exercise prices. Assume the exercise price for call option 1 is 25 and that for call option 2 is 30. The right to buy at a lower price of 25 will be more valuable than the right to buy at a higher price of 30.

Similarly, in the case of a put option, the right to sell for a higher price is more valuable than the right to sell for a lower price.

Relationship between the value of the option and the exercise price

- The value of a European call option is inversely related to the exercise price.
- The value of a European put option is directly related to the exercise price.

Time to Expiration

Longer-term options should be worth more than shorter-term options. For instance, you have a bought a call option on a stock at an exercise price of $25. Which one would be more valuable to you; an option that expires in a week or an option that expires a year later? Without doubt, the option that expires a year later as it gives enough time for the stock price to increase and make the option valuable.

But, is it true for a put option? Partly, yes. The more the time to expiration, the more the opportunity for the stock to fall below its exercise price. However, because put option buyers must wait for the underlying to be sold in order to receive ($X - S_T$) upon exercise, the present value of the payoff decreases as the time to expiry increases. While less common, a longer time to expiration will reduce the value of a put in some cases, particularly for deep-in-the-money puts with a longer time to expiration and a higher risk-free rate of interest.

Relationship between the value of the option and the time to expiration

- The value of a call option is directly related to the time to expiration.
- The value of a put option can be directly or inversely related to the time to expiration. Inversely related under these conditions: the option is deep in the money, and the risk-free rate is high.

Risk-Free Interest Rate

As seen earlier, the exercise values of call and put options are computed as:

Call Option Exercise Value: Max (0, (S_t – PV(X)))

Put Option Exercise Value: Max (0, (PV(X) – S_t))

A higher risk-free rate lowers the PV(X). Therefore, a higher risk-free rate will increase the exercise value of a call option and decrease the exercise value of a put option.

Instructor's Note: The risk-free rate does not directly affect the time value of an option.

Relationship between the value of the option and the risk-free interest rate

- The value of a call option is directly related to the risk-free interest rate.
- The value of a put option is inversely related to the risk-free interest rate.

Volatility of the Underlying

Volatility is good for both call and put options. If the underlying stock becomes more volatile, then the probability of the options expiring deep in the money becomes greater.

Relationship between the value of the option and the volatility of the underlying

- The value of a call option is directly related to the volatility of the underlying.
- The value of a put option is directly related to the volatility of the underlying.

Income or Cost Related to Owning Underlying Asset

		Call option	**Put option**
Benefits (dividends on stocks, interest on bonds, convenience yield on commodities)	Stocks and bonds fall in value when dividends and interest are paid.	Decreases because call holders do not get these benefits.	Increases
Carrying costs		Increases as call holders do not incur these costs.	Decreases

Relationship between the value of the option and benefits/costs of the underlying

- A call option is worth less the more benefits that are paid by the underlying. The call option is worth more the more the costs that are incurred in holding the underlying.
- A put option is worth more the more benefits that are paid by the underlying. The put option is worth less the more costs that are incurred in holding the underlying.

A summary of these factors is presented below.

Increase in	**Value of call option will**	**Value of put option will**
Value of the underlying	Increase	Decrease
Exercise price	Decrease	Increase
Risk-free rate	Increase	Decrease
Time to expiration	Increase	Increase or Decrease
Volatility of the underlying	Increase	Increase
Costs incurred while holding the underlying	Increase	Decrease
Benefits received while holding the underlying	Decrease	Increase

Summary

LO: Explain the exercise value, moneyness, and time value of an option.

At any time before expiration (t < T), an option's value is made up of its exercise value and its time value. The exercise value is the option's value if it could be exercised immediately, whereas the time value captures the possibility that the passage of time and the volatility of the underlying price will increase the profitability of exercise at maturity.

$$\text{Call Option Exercise Value} = \max\left(0, S_t - \frac{X}{(1+r)^{(T-t)}}\right)$$

$$\text{Put Option Exercise Value} = \max\left(0, \frac{X}{(1+r)^{(T-t)}} - S_t\right)$$

The time value of an option is the amount by which the option premium exceeds the exercise value. The longer the time to expiration the higher the time value. The time value of an option is always positive but it declines to zero at maturity.

An option's moneyness expresses the relationship between the underlying price and the exercise price. It refers to whether an option would result in a profit or loss if exercised immediately.

The table below shows the moneyness of an option for various relative values of S and X.

	Call Option	**Put Option**
In the money	S > X	S < X
At the money	S = X	S = X
Out of the money	S < X	S > X

LO: Contrast the use of arbitrage and replication concepts in pricing forward commitments and contingent claims.

Arbitrage:

Due to the conditional nature of payoffs, options have a no-arbitrage upper and lower price bounds.

$$\text{Call option lower bound} = \max\left(0, S_t - \frac{X}{(1+r)^{(T-t)}}\right)$$

$$\text{Call option upper bound} = S_t$$

$$\text{Put Option Lower Bound} = \max\left(0, \frac{X}{(1+r)^{(T-t)}} - S_t\right)$$

$$\text{Put Option Upper Bound} = X$$

Replication:

To replicate a call option at contract inception (t=0), we must borrow at the risk-free rate, r, and use the proceeds to purchase the underlying at a price of S_0. The amount to be borrowed is a proportion of $\frac{X}{(1+r)^T}$ and is based on the likelihood of exercise. As the likelihood of exercise changes the replicating transaction has to be adjusted.

To replicate a put option at contract inception (t=0), we must sell the underlying short at a price of S_0 and lend the proceeds at the risk-free rate, r. The amount to be lent is a proportion of $\frac{X}{(1+r)^T}$ and is based on the likelihood of exercise. As the likelihood of exercise changes the replicating transaction has to be adjusted.

LO: Identify the factors that determine the value of an option and describe how each factor affects the value of an option.

Increase in	**Value of call option will**	**Value of put option will**
Value of the underlying	Increase	Decrease
Exercise price	Decrease	Increase
Risk-free rate	Increase	Decrease
Time to expiration	Increase	Increase or Decrease
Volatility of the underlying	Increase	Increase
Costs incurred while holding the underlying	Increase	Decrease
Benefits received while holding the underlying	Decrease	Increase

LM09 Option Replication Using Put–Call Parity

Version 1.0

1. Introduction

This learning module covers:

- Put-call parity for European options
- Put-call forward parity for European options
- Put-call parity application to explain firm value

2. Put–Call Parity

The four instruments in a put-call parity are as follows:

- Asset: An asset such as a stock with no carrying costs and no interim cash flows. At time 0, the price of the stock is S_0.
- Zero-coupon bond: A zero-coupon bond with a face value of X that matures at time T. At time 0, the value of this bond is $\frac{X}{(1+r)^T}$.
- Call option: A call option on the stock with a strike price of X that expires at time T. S_T is the underlying price at expiration T.
- Put option: A put option on the stock with a strike price of X that expires at time T.

Note: The strike price, X, of the options is the same as the par value of the bond.

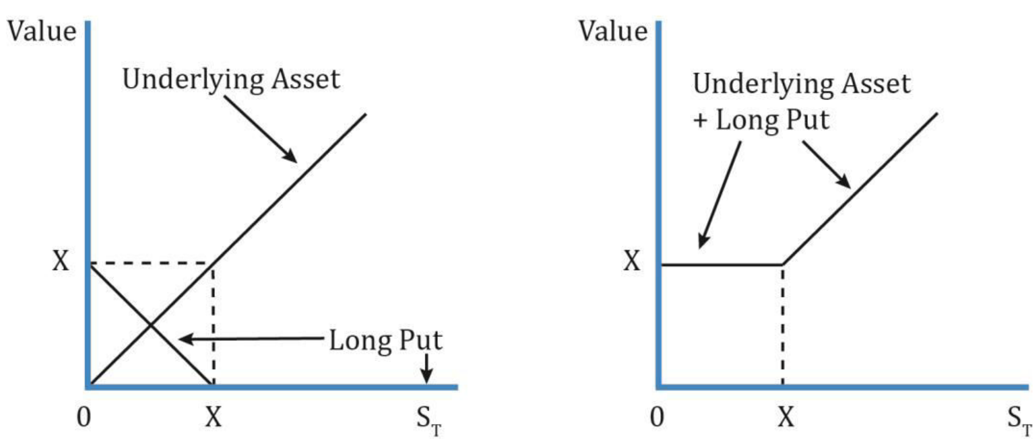
According to put-call parity, fiduciary call = protective put.

Protective Put:

A protective put is to buy the underlying and buy a put option on it. The cost of this strategy is:

$p_0 + S_0$

A protective put is like buying insurance for an asset you own, specifically a stock, to protect against a downside. For instance, you own 1,000 shares of Apple, the market is highly volatile and you are worried about a huge decline in the near term. To limit the downside risk, you can buy a put option on the stock by paying a premium. This is called a protective put.



<u>Interpretation</u>:

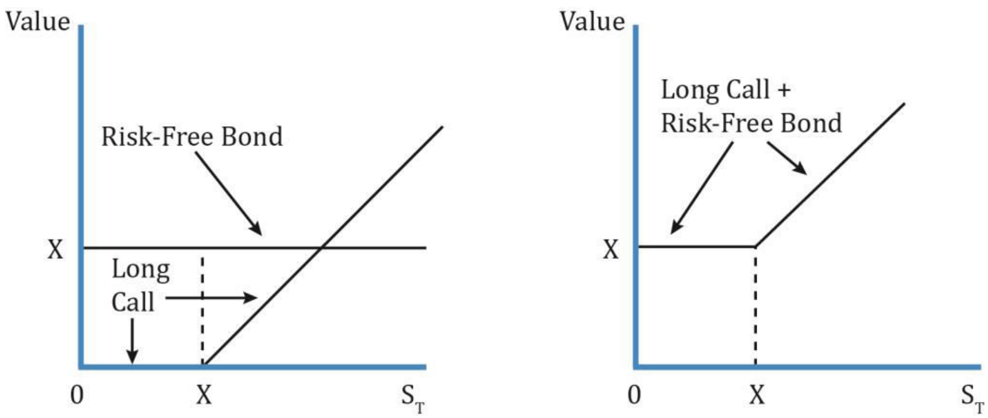
- As you can see, the downside risk is limited to the premium paid for the put. If the underlying price falls below X, it can be still be sold at X by exercising the put option.
- If the underlying rises above X, then it can be sold at the market price and the put option expires worthless.

Fiduciary Call:

A fiduciary call is a long call plus a risk-free bond. The cost of this strategy is:

$$c_0 + \frac{X}{(1+r)^T}$$

The diagram below shows the payoff for a fiduciary call:



Therefore, according to put-call parity:

$$c_0 + \frac{X}{(1+r)^T} = p_0 + S_0$$

In other words, under put–call parity, at t = 0 the price of the long call plus the risk-free asset must equal the price of the long underlying asset plus the long put.

The table below shows how the fiduciary call is equal to the protective put under two possible scenarios: when the stock is above the exercise price (call is in the money) and the stock is below the exercise price (put is in the money).

Outcome at time T when: →	**Put expires in the money ($S_T < X$)**	**Call expires in the money ($S_T >= X$)**
Protective put		
Asset	S_T	S_T
Long puts	$X-S_T$	0
Total	X	S_T
Fiduciary call		
Long call	0	$S_T - X$
Risk-free bond	X	X
Total	X	S_T

	Fiduciary Call	**Protective Put**
Constituents	Long call + risk-free bond	Long put + stock
Equation	$c_0 + \frac{X}{(1+r)^T}$	$p_0 + S_0$
Payoff at T if call expires in the money ($S_T >= X$)	S_T	S_T
Payoff at T if put expires in the money ($S_T < X$)	X	X

If, at time 0, the fiduciary call is not priced the same as the protective put, then there is an arbitrage opportunity.

Example: Put-Call Parity
(*This is based on Example 1 from the curriculum.*)

A stock currently trades at INR295 per share. An investor is considering the purchase of a six-month put on this stock at an exercise price of INR265. A six-month call option with the same exercise price trades in the market at INR59. What should the investor expect to pay for the put if the relevant risk-free rate is 4%?

Solution:

According to put call parity:

$$c_0 + \frac{X}{(1+r)^T} = p_0 + S_0$$

$$59 + \frac{265}{(1+0.04)^{0.5}} = p_0 + 295$$

$$59 + 259.85 = p_0 + 295$$

$$p_0 = \text{INR}23.85$$

The investor should expect to pay a six-month put option premium of p_0 = INR23.85

3. Option Strategies Based on Put–Call Parity

The put-call parity relationship can be rearranged in the following ways:

Synthetic call: $c_0 = p_0 + S_0 - \frac{X}{(1+r)^T}$

Synthetic bond: $\frac{X}{(1+r)^T} = p_0 + S_0 - c_0$

Synthetic stock: $S_0 = c_0 + \frac{X}{(1+r)^T} - p_0$

Synthetic put: $p_0 = c_0 + \frac{X}{(1+r)^T} - S_0$

These equations allow us to replicate an instrument by using the other three instruments.

For example, a put option can be replicated by a combination of a long call, a long risk-free bond and a short position in the underlying.

We can also compare the price of an actual instrument to its synthetic version. If the two prices are different, we can generate riskless profits by exploiting the mispricing.

4. Put–Call Forward Parity and Option Applications

In earlier readings we saw that:

Long asset + Short forward = Long risk-free bond

i.e. Asset – Forward = Risk-free Bond
(Long positions are shown as +ve and short positions as -ve)

Rearranging, we get:

Asset = Forward + Risk-free bond

We will use this relationship to come up with the put-call forward parity.

5. Put–Call Forward Parity

Protective put: Asset + Put

If we substitute the ‘asset’ part in a protective put with a forward and risk-free bond we get:

Synthetic protective put = Forward + Risk-free bond + Put

Here, the risk-free bond has a par value equal to the forward price of $F_0\,(T)$.

The cost this synthetic protective put is:

$$0 + \frac{F_0\,(T)}{(1+r)^T} + p_0$$

The cost of entering a forward contract at t=0 is zero as it requires no upfront payment.

The risk-free bond with par value $F_0\,(T)$ can be purchased at t=0 for $\frac{F_0\,(T)}{(1+r)^T}$

The put option premium at t=0 is p_0

The table below shows that the payoff for a protective put with an asset and a synthetic protective put with a forward contract is the same when the put expires in and out of the money.

	Outcome at T	
	Put Expires In the Money ($S_T < X$)	**Put Expires Out of the Money ($S_T \geq X$)**
Protective put with asset		
Asset	S_T	S_T
Long put	$X - S_T$	0

Total	X	S_T
Protective put with forward contract		
Risk-free bond	$F_0\,(T)$	$F_0\,(T)$
Forward contract	$S_T - F_0\,(T)$	$S_T - F_0\,(T)$
Long put	X - S_T	0
Total	X	S_T

Interpretation:

- Value of the forward contract at expiration = S_T – F_0 (T).
- Notice that the value of the forward contract at expiration is the same irrespective of whether the put expires in or out of the money.

The put-call parity relationship can now be written as:

Synthetic protective put = Fiduciary call

Long Put + Long forward + Risk-free bond = Bond + Call

$$\frac{F_0\,(T)}{(1+r)^T} + p_0 = c_0 + \frac{X}{(1+r)^T}$$

The table below shows that the payoffs for fiduciary call is equal to synthetic protective put when call and put expire in the money.

	Outcome at T	
	Put Expires In the Money ($S_T < X$)	**Put Expires Out of the Money ($S_T \geq X$)**
Synthetic Protective Put		
Risk-free bond	$F_0\,(T)$	$F_0\,(T)$
Forward contract	$S_T - F_0\,(T)$	$S_T - F_0\,(T)$
Long put	X - S_T	0
Total	X	S_T
Fiduciary call		
Call	0	$S_T - X$
Risk-free bond	X	X
Total	X	S_T

6. Option Put–Call Parity Applications: Firm Value

The put-call parity relationship can also be used to model the value of a firm to equity and debt holders.

Assume that at time t = 0, a firm with a market value of V_0 has access to borrowed capital in the form of zero-coupon debt with a face value of D. The value of equity at t = 0 is E_0.

The firm value can be expressed as:

$V_0 = E_0 + PV(D)$

When the debt matures at T, there are two possible outcomes:

1. Solvency ($V_T > D$): The firm value exceeds the face value of debt and we say that the firm is solvent. In this case:
   - Debtholders are paid in full and receive D.
   - Shareholders receive the residual: $E_T = V_T – D$.
2. Insolvency ($V_T < D$): The firm value is below the face value of debt and we say that the firm is insolvent. In this case:
   - Debtholders have a priority claim on assets and receive V_T (which is less than D).
   - Shareholders don't receive any funds, $E_T = 0$.

Thus,

Shareholder payoff is $\max(0, V_T – D)$ and

Debtholder payoff is $\min(V_T, D)$.

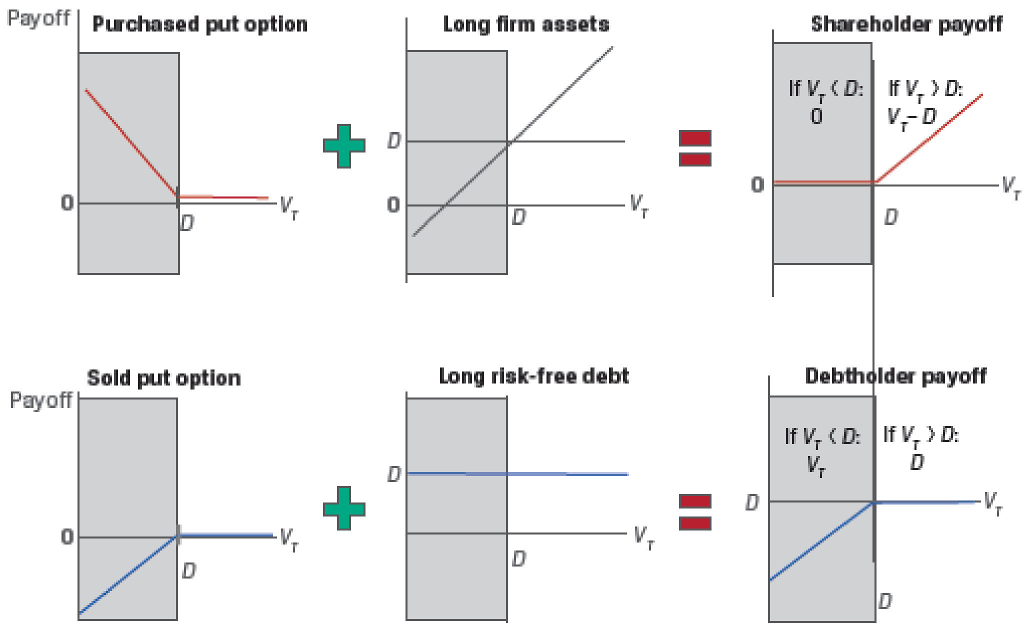
These payoff profiles can be expressed in terms of options:

- Shareholders hold a long position in the underlying firm's assets (V_T) and have purchased a put option on firm value (V_T) with an exercise price of D; that is, $\max(0, D – V_T)$.

  Or we can also say that shareholders have purchased a call option on the firm's assets with an exercise price of D.

- Debtholders hold a long position in a risk-free bond (D) and have sold a put option to shareholders on firm value (V_T) with an exercise price of D.

Exhibit 9 from the curriculum shows the payoff profiles for shareholders and debtholders.



Revisiting the put-call parity relationship:

$S_0 + p_0 = c_0 + PV(X)$

Replacing 'S_0' with 'V_0' and 'X' with 'D' we get:

$V_0 + p_0 = c_0 + PV(D)$

$V_0 = c_0 + PV(D) - p_0$

Thus, the value of the firm is the value to the shareholders (represented by the call option c_0) plus the value to debt holders (represented by the risk-free debt and a sold put option p_0).

Summary

LO: Explain put–call parity for European options.

According to put-call parity, fiduciary call = protective put.

Long call + Risk-free bond = Long put + Stock

$c_0 + \frac{X}{(1+r)^T} = p_0 + S_0$

LO. Explain put–call forward parity for European options.

According to put-call parity, since a fiduciary call = protective put, a fiduciary call must also be equal to a synthetic protective created using a forward contract and a risk-free bond.

Fiduciary call = Synthetic protective put

Long call + Risk-free bond = Long Put + Long forward + Risk-free bond

$$c_0 + \frac{X}{(1+r)^T} = p_0 + \frac{F_0\,(T)}{(1+r)^T}$$

The put-call parity relationship can also be applied to explain the value of a firm:

$V_0 = c_0 + PV(D) - p_0$

The value of the firm is the value to the shareholders (represented by the call option c_0) plus the value to debt holders (represented by the risk-free debt and a sold put option p_0).

LM10 Valuing a Derivative Using a One-Period Binomial Model

Version 1.0

1. Introduction

This learning module covers:

- How to value a derivative using a one-period binomial model
- The concept of risk-neutrality in derivatives pricing

2. Binomial Valuation

According to the law of one price, if the payoffs from any two assets at a given future date are identical, then the value of these two assets must also be identical today.

Forward commitments can be priced without making assumptions about the future price of the underlying asset. However, due to their asymmetric payoffs, the pricing of options requires a model to factor in the random price behavior of the underlying asset.

A commonly used model to determine the no-arbitrage value of an option is the binomial model.

3. The Binomial Model

The binomial model assumes that over a given time period, there are only two possible outcomes - the asset's price will either go up to S_1^u or down to S_1^d.

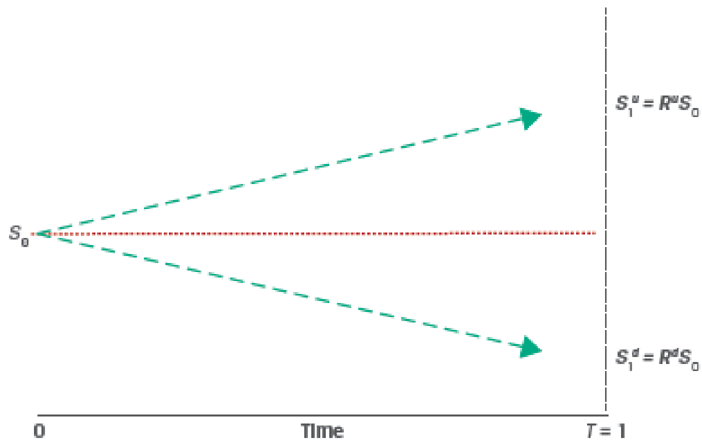
Let q denote the probability of an upward price movement and $1 - q$ denote the probability of a downward price movement. With only two possible outcomes, the sum of these two probabilities must equal to 1.

To value options, knowing q is not required, only the values S_1^u and S_1^d are needed. The difference between S_1^u and S_1^d represents the spread of possible future price outcomes or the volatility of the underlying asset.

We can also define the upward and downward moves in terms of returns

$R^u = S_1^u / S_0 > 1$
$R^d = S_1^d / S_0 < 1$

Exhibit 1 from the curriculum, illustrates the binomial model.



4. Pricing a European Call Option

Consider a one-year European call option with an exercise price (X) of 100.

The underlying spot price (S_0) is 80.

The two possible stock prices after 1 year are: S_1^d = 60 and S_1^u=110

We can compute the up and down returns as:

R^u = 110/80 = 1.375

R^d = 60/80 = 0.75

At $t = 0$, the call option value is c_0.

At $t = 1$, the call option value is either c_1^u (if the price goes up) or c_1^d (if the price goes down). We can compute the values c_1^u and c_1^d as:

Up move: The call option ends in-the-money

$c_1^u = \text{Max}\ (0, \text{S}_T - \text{X}) = \text{Max}(0{,}110 - 100) = 10$

Down move: The call option ends out-of-the-money

$c_1^d = \text{Max}\ (0, \text{S}_T - \text{X}) = \text{Max}(0{,}60 - 100) = 0$

To compute c_0 we can use replication and no-arbitrage pricing. Assume that at $t = 0$ we sell the call option at a price of c_0 and purchase h units of the underlying asset. V_0 represents the initial investment in the portfolio, and V_1^u and V_1^d represent the portfolio value if the underlying price moves up or down, respectively.

$V_0 = hS_0 - c_0$

$V_1^u = hS_1^u - c_1^u$

$V_1^d = hS_1^d - c_1^d$

We solve for h, such that the portfolio payoff in both the up and down scenarios are equal, i.e.:

$$V_1^u = V_1^d$$

$$hS_1^u - c_1^u = hS_1^d - c_1^d$$

Solving for h, we get:

$$h = \frac{c_1^u - c_1^d}{S_1^u - S_1^d}$$

This equation gives us the hedge ratio, or the proportion of the underlying that will offset the risk associated with an option.

The hedge ratio for our example is:

$$h = \frac{c_1^u - c_1^d}{S_1^u - S_1^d} = \frac{10 - 0}{110 - 60} = 0.20$$

i.e. for each call option unit sold, we must buy 0.2 units of the underlying asset.

The portfolio values at t=1 for the up and down scenarios are:

$$V_1^u = hS_1^u - c_1^u = 0.20 \times 110 - 10 = 12$$

$$V_1^d = hS_1^d - c_1^d = 0.20 \times 60 - 0 = 12$$

As expected, the portfolio values are the same, which has two implications:

1. We can use either portfolio to value the derivative, and
2. The return $V_1^u/V_0 = V_1^d/V_0$ must equal one plus the risk-free rate.

To prevent arbitrage, the portfolio value at *t =1* should be discounted at the risk-free rate so that:

$$V_0 = \frac{V_1}{1 + r}$$

For our example, V_1 =12. Assume that the annual risk-free rate is 5%. Therefore,

$$V_0 = \frac{12}{1.05} = 11.42$$

Replacing V_0 by the hedged portfolio we get:

$$hS_0 - c_0 = 11.42$$

$$c_0 = hS_0 - 11.42 = 0.2 \times 80 - 11.42 = 4.58$$

The value of the call option at *t = 0* is 4.58.

Example:

(*This is based on Example 1 from the curriculum.*)

Hightest Capital believes that a particular non-dividend-paying stock is currently trading at \$50 and is considering the sale of a one-year European call option at an exercise price of \$55.

If the stock price is expected to either go up or down by 20% over the next year, what price should Hightest expect to receive for the sold call option? Assume a risk-free rate of 5%.

Solution:

$S_0 = 50$

$S_1^u = R^u S_0 = 1.2 \times 50 = 60$

$S_1^d = R^d S_0 = 0.8 \times 50 = 40$

The value of the call option at expiry is:

$c_1^u = \text{Max}\ (0, \text{S}_T - \text{X}) = \text{Max}(0{,}60 - 55) = 5$

$c_1^d = \text{Max}\ (0, \text{S}_T - \text{X}) = \text{Max}(0{,}40 - 55) = 0$

The hedge ratio of the option is:

$$h = \frac{c_1^u - c_1^d}{S_1^u - S_1^d} = \frac{5 - 0}{60 - 40} = 0.25$$

The value of the hedged portfolio at maturity is:

$V_1 = V_1^u = hS_1^u - c_1^u = 0.25 \times 60 - 5 = 10$

The present value of the hedged position today is:

$V_0 = V_1(1 + r)^{-1} = \$10.00 \times 1.05^{-1} = 9.52$

The call option value is:

$c_0 = h^*S_0 - V_0 = 0.25 \times 50.00 - 9.52 = 2.98$

5. Risk Neutrality

As seen in the previous section, neither the actual (real-world) probabilities of underlying price increases or decreases nor the expected return of the underlying are required to price an option. Only the expected volatility i.e. the values S_1^u and S_1^d are required to price an option.

Another method of calculating the value of a call option today, c_0, is as the expected value of the option at expiration, c_1^u and c_1^d, discounted at the risk-free rate, *r*.

$$c_0 = \frac{(\pi c_1^u + (1-\pi)c_1^d)}{(1+r)^T}$$

In this equation π and (1-π) represent **risk-neutral probabilities** (rather than actual probabilities) of an up move and down move respectively.

The risk-neutral probability can be calculated as:

$$\pi = \frac{1 + r - R^d}{R^u - R^d}$$

Substituting the details from our earlier example: R^u = 1.375, R^d = 0.75, c_1^u= 10, c_1^d=0 and assuming an annual risk-free rate of 5%, we get:

$$\pi = \frac{1 + r - R^d}{R^u - R^d} = \frac{1 + 0.05 - 0.75}{1.375 - 0.75} = 0.48$$

We can compute the value of c_0 as:

$$c_0 = \frac{(\pi c_1^u + (1-\pi)c_1^d)}{(1+r)^T} = \frac{0.48 \times 10 + 0.52 \times 0}{1.05} = 4.57$$

This matches our result from the previous section.

The risk-neutral probabilities used in this method are determined solely by the up and down gross returns, R^u and R^d, representing underlying asset volatility and the risk-free rate. The investor view on risk is not considered hence this method is referred to as **risk-neutral pricing**.

Example:
(*This is based on Example 2 from the curriculum.*)

Using the same details as Example 1:

"Hightest Capital believes that a particular non-dividend-paying stock is currently trading at $50 and is considering the sale of a one-year European call option at an exercise price of $55.

If the stock price is expected to either go up or down by 20% over the next year, what price should Hightest expect to receive for the sold call option? Assume a risk-free rate of 5%."

1. Compute the risk-neutral probability of an up move and down move.

2. Calculate the no-arbitrage price of the call option.

Solution to 1:

The risk neutral probability of an up move is:

$$\pi = \frac{1 + r - R^d}{R^u - R^d} = \frac{1 + 0.05 - 0.8}{1.2 - 0.8} = 0.625$$

The risk-neutral probability of a down move is therefore:

1 – π = 1 – 0.625 = 0.375.

Solution to 2:

$$c_0 = \frac{(\pi c_1^u + (1-\pi)c_1^d)}{(1+r)^T}$$

$$c_0 = \frac{0.625 \times Max(0{,}60-55) + 0.52 \times Max(0{,}40-55)}{1.05} = \frac{3.125}{1.05} = 2.98$$

Summary

LO. Explain how to value a derivative using a one-period binomial model.

The binomial model assumes that over a given time period, there are only two possible outcomes - the asset's price will either go up to S_1^u or down to S_1^d.

The hedge ratio for a call option is calculated as:

$$h = \frac{c_1^u - c_1^d}{S_1^u - S_1^d}$$

To compute c_0 we can use replication and no-arbitrage pricing. Assume that at $t = 0$ we sell the call option at a price of c_0 and purchase h units of the underlying asset.

$$V_0 = hS_0 - c_0$$

$$V_1^u = hS_1^u - c_1^u$$

$$V_1^d = hS_1^d - c_1^d$$

To prevent arbitrage, the portfolio value at $t = 1$ should be discounted at the risk-free rate so that:

$$V_0 = \frac{V_1}{1 + r}$$

or

$$hS_0 - c_0 = \frac{V_1}{1 + r}$$

LO. Describe the concept of risk neutrality in derivatives pricing.

Another method of calculating the value of a call option today, c_0, is as the expected value of the option at expiration, c_1^u and c_1^d, discounted at the risk-free rate, *r*.

$$c_0 = \frac{(\pi c_1^u + (1 - \pi)c_1^d)}{(1 + r)^T}$$

The risk-neutral probability can be calculated as:

$$\pi = \frac{1 + r - R^d}{R^u - R^d}$$

The risk-neutral probabilities used in this method are determined solely by the up and down gross returns, R^u and R^d, representing underlying asset volatility and the risk-free rate. The investor view on risk is not considered hence this method is referred to as risk-neutral pricing.

